Emergent Specialization in Learner Populations: Reward-Independent Capacity Assignment as a Defense Against Catastrophic Forgetting

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Abstract

Capacity-limited learners serving many tasks face an allocation problem: with finite per-learner capacity, which learner handles which task—and who *keeps* a task when it falls dormant? The standard machine-learning answer is a learned gating network (mixture-of-experts), and recent theory shows MoE can mitigate catastrophic forgetting in continual learning—provided experts are plentiful and the gate's updates eventually stop Li et al. (2025). We study the complementary regime where those provisos fail—**fixed total capacity, experts forced into reuse ($K \ll R$), gate still learning**—and find a sharp dissociation governed by a single property: whether capacity assignment is **independent of current task reward**. Our instrument is **NichePopulation**, a deliberately minimal mechanism coupling winner-take-all competition with the canonical exponentiated-gradient (Hedge / multiplicative-weights) niche update. With no communication, shared reward, gate, or diversity objective, the population self-organizes into one-per-regime specialists (competition alone, no niche bonus, suffices: mean SI $= 0.65$, every domain > 0.49). For **spatial coverage**, this emergent division of labor matches purpose-built alternatives—an EOI/CDS-style diversity objective and a learned MoE router—so its value there is parsimony (synthetic +57%, traffic +8.7% at $K{=}1$ over an equal-capacity random-diversity population; no gate or auxiliary objective needed). For **retention under non-stationarity**, the mechanisms split exactly along the reward-independence line. A reward-driven gate receives *no signal* from a dormant regime—the one thing needing protection is the one thing emitting no reward—so it reallocates the expert and relearns on reactivation, like the capacity-bounded monolith (+71% lower post-reactivation error for the population, $p \sim 10^{-35}$; robust to replacing discrete eviction with soft interference). Every arm whose assignment ignores current task reward (fixed random niches, intrinsic-reward diversity, converged competitive exclusion) retains; every arm whose allocation chases it (monolith, router) forgets. This population-level account independently corroborates the theoretical finding that MoE gates must be frozen for continual-learning convergence—competition reaches the required reward-independent assignment *emergently*, with no freezing schedule to tune. We pair these results with an honest audit: the Specialization Index certifies *organization*, not exploitable structure; with slack capacity a regime-conditioned monolith matches the population, so division of labor pays **specifically** under bounded capacity and non-stationarity. Cooperative MARL baselines (IQL, VDN, QMIX, MAPPO) homogenize (SI ≤ 0.02). Validated on six real-world domains (crypto, commodities, weather, solar, traffic, air quality; 145,294 records). We position competitive exclusion as the most **parsimonious** known route to reward-independent specialization—competitive with explicit diversity objectives, and structurally immune to a failure mode of reward-driven routing. Code and data: `https://github.com/HowardLiYH/NichePopulation`.

1 Introduction

Many learning systems must serve a task space larger than any single model can hold: finite memory, parameters, or attention against many distinct regimes whose relevance shifts over time. The allocation question—*which learner handles which regime, and who retains a regime that falls dormant?*—has a standard answer: learn a router. Mixture-of-experts architectures gate each input to a capacity-limited expert, and a growing continual-learning literature credits exactly this isolation with mitigating catastrophic forgetting Li et al. (2025). But the theory carries two provisos that are easy to miss: the guarantees hold when experts are plentiful relative to tasks (or can be added), and—strikingly—when the gating network's updates are eventually *terminated*, because a gate that keeps adapting under task arrival destabilizes the assignment Li et al. (2025). Both provisos point at the same underlying property, which this paper isolates: **retention requires capacity assignment that is independent of current task reward.**

We demonstrate this dissociation in a controlled population-learning setting and exhibit a mechanism that satisfies the property *for free*. When per-agent capacity is bounded ($K \ll R$) and regimes cycle through dormancy, every arm in our study whose assignment ignores current task reward retains dormant regimes—fixed random niches, an EOI/CDS-style intrinsic-diversity objective, and emergent competitive exclusion—while every arm whose allocation chases current reward forgets and relearns: the capacity-bounded monolith *and* the learned MoE router (+71% lower post-reactivation error for the competitive population, $p \sim 10^{-35}$; the separation survives replacing discrete eviction with graded interference). The router's failure has a one-line cause we make precise (Observation 1): a dormant regime emits no reward gradient, so a reward-driven gate has no signal with which to reserve capacity for it. Routing and protection draw on the same signal, and the regime most needing protection is the one producing none.

This challenge manifests across numerous domains with significant real-world impact. In autonomous driving, vehicles must implicitly coordinate traffic flow to avoid congestion and collisions. In algorithmic trading, strategies must avoid correlated behaviors that amplify market volatility—the 2010 Flash Crash, where the Dow Jones dropped 1,000 points in minutes, exemplifies the catastrophic consequences of homogeneous learner behaviors Kirilenko et al. (2017). In distributed sensing networks, sensors must specialize to different environmental conditions to maximize information gain.

Existing approaches to coordinating learner populations typically require one of three mechanisms: (1) **explicit communication channels** that enable learners to share intentions and negotiate roles Foerster et al. (2016), (2) **centralized training** with shared reward functions that encourage cooperative behaviors Lowe et al. (2017), or (3) **handcrafted diversity incentives** such as quality-diversity archives that explicitly maintain behavioral variety Pugh et al. (2016); Mouret and Clune (2015). While these methods have demonstrated success in specific settings, they introduce significant complexity and may not scale to large, decentralized systems where communication is costly or impossible.

Among reward-independent mechanisms, why single out *competition*? Because it is the most parsimonious: random fixed niches retain but cannot guarantee coverage at K=1; explicit diversity objectives retain but require an auxiliary intrinsic reward; gate-freezing retains but requires deciding *when* to freeze. Competition needs none of these. The intuition is ecological—competitive exclusion drives species with identical niches apart Hardin (1960), as with Darwin's finches, which diversified their beaks under competition for food rather than by communicating Grant and Grant (2014). When learners compete for limited rewards in regime-switching environments, homogeneous strategies become unstable: learners with identical behaviors compete directly, reducing their expected payoffs, so deviation to a less-contested niche becomes profitable. Our NichePopulation algorithm operationalizes this pressure, and—crucially for retention—the converged assignment is *structural*: a specialist simply idles while its niche is dormant, retaining it with no signal required. Competition itself thus serves as a coordination mechanism, and the coordination it produces is precisely of the reward-independent kind that retention demands.

Crucially, we demonstrate that **strict competitive exclusion (winner-take-all dynamics) is not merely a simplification but the key structural driver of differentiation**: softer competition allows learners to "hedge" across niches, destabilizing the clean partition we seek. We prove this formally in Proposition 1, showing that under winner-take-all dynamics, homogeneous strategies are not Nash equilibria. This stands in contrast to standard multi-agent reinforcement learning (MARL) methods, which optimize shared value functions or critics, inadvertently driving learners toward behavioral homogeneity rather than diversity.

To test this hypothesis, we introduce **NichePopulation**, a deliberately simple algorithm that induces emergent specialization through three mechanisms:

1. **Competitive exclusion**: Only the best-performing learner in each iteration receives positive updates, creating winner-take-all dynamics that penalize homogeneous behaviors.

2. **Niche affinity tracking**: Learners maintain probability distributions over environmental regimes, developing preferences for conditions where they consistently perform well.

3. **Optional niche bonus**: Learners receive amplified rewards when operating in their preferred regime, controlled by parameter $\lambda \geq 0$.

Critically, we demonstrate that **setting $\lambda = 0$ (no niche bonus) still produces significant specialization**—the niche bonus accelerates specialization but is not its cause. This validates our ecological hypothesis: diversity emerges from competitive dynamics alone, without explicit incentives.

Contributions. Our work makes four contributions to the literature on learner populations, emergent coordination, and continual learning:

1. **A reward-independence account of retention under bounded capacity (our central result).** Across five capacity-allocation mechanisms—monolith, learned MoE router, fixed random niches, an EOI/CDS-style diversity objective, and emergent competitive exclusion—we show that retention of dormant regimes tracks a single property: whether capacity assignment is independent of current task reward. Reward-chasing arms (monolith, router) forget and relearn on reactivation; reward-independent arms retain (+71% lower post-reactivation error, $p \sim 10^{-35}$; robust to swapping LRU eviction for soft interference). An idealized Observation explains why—dormant regimes emit no protective reward signal—and independently corroborates, from population dynamics, the theoretical finding that MoE gates must be frozen for continual-learning convergence Li et al. (2025).

2. **When emergent specialization improves accuracy.** Controlled ablations localize *when* division of labor is a task-accuracy lever. With slack capacity it is not—a single regime-conditioned monolith matches the population. Under (i) *bounded per-agent capacity* ($K \ll R$) and (ii) *non-stationarity with finite memory*, competitive exclusion contributes a genuine, well-controlled advantage: *guaranteed coverage* (synthetic +57%, traffic +8.7% at K=1 vs. an equal-capacity random-diversity population) and *distributed persistent memory* (+71% post-reactivation vs. a capacity-matched monolith)—with parsimony, not dominance, over purpose-built diversity and routing baselines for spatial coverage.

3. **A coordination-free self-organization mechanism.** We introduce NichePopulation, which couples winner-take-all competition with an exponentiated-gradient niche update to make a population partition into one-per-regime specialists *without* communication, shared rewards, gates, or diversity incentives—competition alone ($\lambda = 0$) suffices (mean SI = 0.65 at $\lambda = 0$, every domain > 0.49). The converged assignment is structural—specialists idle through dormancy—which is exactly the reward-independent property that retention requires.

4. **An honest meaningfulness audit and MARL contrast.** We show the Specialization Index measures *organization*, not structure-dependence: it is invariant to regime-label shuffling and inflatable by the learning rate. We add an oracle structure-diagnostic, causal (leakage-free) regime labels, and coupling/monolith ablations that separate regime-*conditioning* from population *specialization*. Standard cooperative MARL baselines (IQL, VDN, QMIX, MAPPO) homogenize under shared objectives (SI ≤ 0.02). Validated on six real-world domains (145,294 records).

2 Related Work

Multi-Agent Reinforcement Learning. MARL methods have achieved remarkable success in cooperative and competitive settings. Independent Q-Learning (IQL) Tan (1993) allows learners to learn independently but struggles with non-stationarity. Value decomposition methods like QMIX Rashid et al. (2018) and

VDN factor joint value functions to enable centralized training with decentralized execution. Multi-Agent PPO (MAPPO) Yu et al. (2022) extends policy gradient methods to multi-learner settings with shared critics. These *shared-objective, parameter-sharing* methods optimize task return rather than learner diversity, and a well-known consequence is behavioral homogenization Li et al. (2021): as we confirm empirically, off-the-shelf cooperative MARL converges to near-identical value functions (specialization index below 0.02) despite heterogeneous environments. We stress that these are the appropriate *cooperative* baselines, not the appropriate *diversity* baselines—they do not try to specialize. A separate line of *diversity- and role-emergence MARL* does target heterogeneity: ROMA Wang et al. (2020) and RODE Wang et al. (2021) learn role embeddings/decompositions, MAVEN Mahajan et al. (2019) injects diverse exploration, and EOI Jiang and Lu (2021) and CDS Li et al. (2021) add intrinsic individuality/mutual-information objectives, with R3DM Goel et al. (2025) the current state of the art. These methods produce the kind of division of labor we study, but require an explicit diversity objective, a learned classifier or dynamics model, and a deep policy network. Our contribution is orthogonal and complementary: we show that the *same* coverage arises from a lightweight competitive rule with no diversity objective, and—to avoid a strawman—we benchmark our bounded-capacity claims against an explicit EOI/CDS-style diversity objective directly (Section 5.8), finding it a strong baseline that competition matches but does not dominate.

Quality-Diversity Optimization. The quality-diversity (QD) paradigm explicitly optimizes for both performance and behavioral diversity. MAP-Elites Mouret and Clune (2015) maintains an archive of diverse, high-performing solutions indexed by behavior descriptors. Novelty search Lehman and Stanley (2011) rewards behavioral novelty rather than task performance. These methods have proven effective in evolutionary robotics and game playing. However, QD methods require defining behavior descriptors *a priori*—a non-trivial design choice that may miss important dimensions of variation. Our approach achieves diversity without explicit diversity objectives, emerging naturally from competitive dynamics.

Ecological Niche Theory. The competitive exclusion principle, formalized by Gause Gause (1934) and Hardin Hardin (1960), states that complete competitors cannot coexist: species with identical ecological niches will compete until one is driven to extinction or evolves to occupy a different niche. MacArthur's work on resource partitioning MacArthur (1958) showed how species divide resources along multiple dimensions to reduce competition. We formalize these ecological concepts for artificial learners, proving that competitive dynamics in learner populations produce analogous niche partitioning.

Ensemble Methods and Diversity. Ensemble methods combine diverse models to improve prediction accuracy Dietterich (2000). Traditional approaches achieve diversity through random initialization, bagging, or boosting—methods that introduce diversity exogenously. Our work demonstrates that diversity can emerge endogenously through competition, without requiring external mechanisms.

Continual Learning and Catastrophic Forgetting. A capacity-bounded learner trained on a shifting sequence of tasks tends to overwrite earlier knowledge—*catastrophic forgetting* French (1999). The dominant remedies operate within a single model: weight regularization that protects parameters important to old tasks (e.g. elastic weight consolidation Kirkpatrick et al. (2017)), rehearsal, or architectural growth such as progressive networks Rusu et al. (2016); see Parisi et al. (2019) for a review. Our non-stationarity result (Section 5.2) contributes a complementary, *population-level* perspective: when capacity is bounded, an emergent division of labor distributes the task space across specialists so that dormant regimes are retained by idle specialists rather than overwritten—forgetting-resistance as a structural, coordination-free byproduct of competition. Notably, we find a reward-driven Mixture-of-Experts router does *not* achieve this, because optimizing current task reward gives no incentive to reserve capacity for dormant tasks.

Mixture-of-Experts in Continual Learning. A recent line of work argues MoE *mitigates* forgetting by isolating tasks in experts, with theory establishing expert diversification and load balancing in overparameterized linear settings Li et al. (2025)—under two conditions: sufficient experts relative to tasks, and termination of gating-network updates after an exploration phase, without which continual task arrival destabilizes routing Li et al. (2025). Our results complement this picture from the regime the theory excludes:

when total capacity is fixed, experts are forced into reuse ($K \ll R$), and the gate continues to learn, isolation breaks down and the router inherits the monolith's forgetting (Section 5.2). Our Observation 1 gives the mechanism—dormant tasks emit no protective reward signal—and our population-level finding that retention tracks *reward-independence of assignment* recovers, from competitive dynamics rather than regression theory, the same prescription as gate-freezing. The distinction matters in practice: freezing requires a schedule; competitive exclusion converges to a reward-independent assignment emergently.

General vs. Specialized Intelligence. The question of whether intelligence is inherently general or specialized has resurfaced in recent discussions among AI researchers. LeCun argues that human intelligence is "super-specialized for the physical world" and that our perception of generality is illusory—we cannot imagine the problems we are blind to LeCun (2022). He provides a compelling mathematical argument: the space of possible boolean functions on sensory inputs ($2^{2^{10^6}}$) vastly exceeds what any brain can represent ($2^{3.2\times10^{15}}$), making biological intelligence "ridiculously specialized" by necessity. Others counter that this conflates generality with universality: in the Turing Machine sense, brains are capable of learning anything computable given sufficient resources. However, practical systems inevitably require some degree of specialization around the target distribution being learned.

Our work provides a bridge between these perspectives. We show that specialization is not merely a constraint imposed by finite resources, but an *emergent property* of competitive learner populations. Even learners with identical, general-purpose capabilities will spontaneously partition into specialists under competitive pressure (Proposition 1). This suggests that the debate may be missing a key insight: generality and specialization are not opposites—rather, specialization *emerges from* general systems under ecological pressure. The No Free Lunch theorem guarantees that no single strategy can dominate across all conditions; competition enforces this principle by driving learners toward complementary niches.

3 Method

3.1 Problem Formulation

Consider a population of N learners operating in a regime-switching environment. The environment transitions between R distinct regimes $\mathcal{R} = \{r_1, \ldots, r_R\}$, where each regime represents a qualitatively different state with distinct dynamics. In financial markets, regimes might include bull markets, bear markets, and sideways consolidation; in weather prediction, regimes might distinguish clear, cloudy, and stormy conditions.

At each timestep t, the environment is in regime $r_t \in \mathcal{R}$, drawn from stationary distribution $\pi(r)$. Each learner i selects a prediction method $m_i \in \mathcal{M}$ from a shared inventory of M methods. The learner then receives reward $R_i(r_t, m_i)$ based on the method's performance in the current regime. Crucially, different methods have different strengths across regimes: a momentum-following strategy may excel in trending markets but fail during mean-reversion periods.

Definition 1 (Regime-Method Affinity)**.** *The affinity $A(r, m) \in [0, 1]$ measures the expected performance of method m in regime r, normalized such that $\max_m A(r, m) = 1$ for each regime.*

Our goal is to induce **emergent specialization**: learners should spontaneously partition into specialists for different regimes, without explicit coordination or diversity incentives. We measure specialization through an entropy-based index.

Definition 2 (Specialization Index)**.** *For a learner with niche affinity distribution $\alpha \in \Delta^R$ (probability simplex over regimes), the Specialization Index is:*

$$SI(\alpha) = 1 - \frac{H(\alpha)}{\log R} \tag{1}$$

where $H(\alpha) = -\sum_r \alpha_r \log \alpha_r$ is Shannon entropy. SI = 1 indicates perfect specialization (all probability on one regime); SI = 0 indicates uniform distribution (no specialization).

3.2 The NichePopulation Algorithm

Each learner i maintains two learned representations:

- **Method beliefs** $\beta_{i,r,m} \in \mathbb{R}^+$: Parameters of a Beta distribution representing expected performance of method m in regime r. These beliefs are updated through Bayesian inference.
- **Niche affinity** $\alpha_i \in \Delta^R$: A probability distribution over regimes representing the learner's specialization. This distribution evolves based on competitive outcomes.

Method Selection via Thompson Sampling. When the environment is in regime r_t, learner i selects method m by sampling from belief distributions and choosing greedily:

$$m_i = \operatorname*{argmax}_{m \in \mathcal{M}} \tilde{\theta}_m, \quad \text{where } \tilde{\theta}_m \sim \text{Beta}(\beta^+_{i,r_t,m}, \beta^-_{i,r_t,m}) \tag{2}$$

Thompson Sampling provides principled exploration-exploitation balance: uncertain methods are occasionally selected to gather information, while well-understood high-performing methods are exploited.

Competitive Exclusion. All learners execute their selected methods simultaneously and receive rewards. The **winner** is the learner with highest adjusted reward, where the niche bonus is an *additive* shaping term centred on the uniform affinity $1/R$:

$$i^* = \operatorname*{argmax}_{i} \tilde{R}_i, \quad \text{where } \tilde{R}_i = R_i + \lambda \cdot \left(\alpha_{i,r_t} - \tfrac{1}{R}\right) \tag{3}$$

Here $\lambda \geq 0$ is the niche bonus coefficient and α_{i,r_t} is learner i's affinity for the *current* regime r_t. When $\lambda > 0$, a learner that has come to prefer the current regime ($\alpha_{i,r_t} > 1/R$) is advantaged and one that under-weights it is penalised, creating additional pressure for specialization. The $-1/R$ baseline centres the term (it is constant across learners and hence cancels in the argmax); it is retained only to match the implementation. This additive shaping is exactly the rule used in the released code and the companion deep-dive Li (2026); at $\lambda = 0$ it vanishes, so the winner is decided by raw reward alone.

The key mechanism is **competitive exclusion**: only the winner receives positive belief updates. This creates winner-take-all dynamics where learners with identical strategies compete directly, reducing their expected payoffs.

Belief and Affinity Updates. After competition, the winner updates their method beliefs:

$$\beta^+_{i^*,r_t,m_{i^*}} \leftarrow \beta^+_{i^*,r_t,m_{i^*}} + \tilde{R}_{i^*} \tag{4}$$

and their niche affinity using an *exponentiated-gradient* (EG) update on the simplex:

$$\alpha^{(t+1)}_{i^*,r} \;\leftarrow\; \frac{\alpha^{(t)}_{i^*,r} \exp(\eta\, \mathbf{1}[r = r_t])}{\sum_{k=1}^{R} \alpha^{(t)}_{i^*,k} \exp(\eta\, \mathbf{1}[k = r_t])}. \tag{5}$$

The denominator is the standard EG partition function and intrinsically preserves the simplex $\alpha_{i^*} \in \Delta^R$ *by construction*—no post-hoc clamping or rescaling is required. Equation (5) is the multiplicative-weights / Hedge update applied to the regime simplex with a unit reward on the realized regime Kivinen and Warmuth (1997); Freund and Schapire (1997); Arora et al. (2012). In the small-η limit it reduces to the replicator dynamics $\dot{\alpha}_r = \alpha_r(g_r - \bar{g})$ with $g_r = \mathbf{1}[r = r_t]$, providing a clean ecological reading. A full derivation, proofs of correctness, and a worked comparison against the additive heuristic used in V1–V3 of this paper appear in the companion deep-dive Li (2026).

Algorithm 1 NichePopulation

Require: Population of N learners, regimes $\mathcal{R}$, methods $\mathcal{M}$, bonus λ, learning rate η

Initialize $\beta^+_{i,r,m} \leftarrow 1, \beta^-_{i,r,m} \leftarrow 1$ and $\alpha_{i,r} \leftarrow 1/R$ for all i, r, m

for iteration $t = 1, 2, \ldots, T$ **do**

  Sample regime $r_t \sim \pi(r)$

  for each learner $i = 1, \ldots, N$ **do**

    Sample $\tilde{\theta}_m \sim \text{Beta}(\beta^+_{i,r_t,m}, \beta^-_{i,r_t,m})$ for all m

    Select method: $m_i \leftarrow \text{argmax}_m \tilde{\theta}_m$

    Execute method, observe reward R_i

    Compute adjusted reward: $\tilde{R}_i \leftarrow R_i + \lambda \cdot (\alpha_{i,r_t} - 1/R)$

  end for

  Determine winner: $i^* \leftarrow \text{argmax}_i \tilde{R}_i$

  Update winner's method belief: $\beta^+_{i^*,r_t,m_{i^*}} \leftarrow \beta^+_{i^*,r_t,m_{i^*}} + \tilde{R}_{i^*}$

  {Update winner's niche affinity (EG / Hedge)}

  $\alpha_{i^*,r_t} \leftarrow \alpha_{i^*,r_t} \cdot \exp(\eta); \quad Z \leftarrow \sum_k \alpha_{i^*,k}; \quad \alpha_{i^*,k} \leftarrow \alpha_{i^*,k}/Z \, \forall k$

end for

3.3 Method Specialization

Beyond regime specialization, we observe that learners develop preferences for specific prediction methods. We quantify this through:

Definition 3 (Method Specialization Index)**.** *For a learner with method usage distribution $\pi_i \in \Delta^M$, the Method Specialization Index is:*

$$MSI(\pi_i) = 1 - \frac{H(\pi_i)}{\log M} \tag{6}$$

Definition 4 (Method Coverage)**.** *The population's method coverage is the fraction of methods used by at least one specialist:*

$$Coverage = \frac{|\{m : \exists i, \pi_{i,m} > \tau\}|}{M} \tag{7}$$

where τ is a threshold (we use $\tau = 0.3$).

High method coverage indicates division of labor: different learners specialize in different prediction strategies, collectively utilizing the population's full repertoire.

4 Theoretical Analysis

We give two propositions and one heuristic remark that characterize the *mechanism* behind emergent specialization: the instability of homogeneous profiles under competition (Proposition 1) and the necessity of regime heterogeneity (Proposition 2). We are explicit about their status: they are stability/best-response arguments under stylized payoff models and are meant as guides to intuition. Our quantitative claims rest on the controlled experiments of Sections 5–5.8, not on these arguments.

Proposition 1 (Competitive Exclusion)**.** *In a competitive learner population with N learners, R regimes, and winner-take-all dynamics, if two learners i and j have identical niche affinities ($\alpha_i = \alpha_j$), then the strategy profile is not a Nash equilibrium when $N > R$.*

Proof. Let learners i and j share identical affinities $\alpha_i = \alpha_j = \alpha$. In any regime r, both learners select methods from identical belief distributions and compete for the same reward. Under winner-take-all dynamics, the expected payoff for each is:

$$\mathbb{E}[\text{Payoff}_i | r] = \frac{V_r}{k_r} - c \tag{8}$$

where V_r is the regime value, k_r is the number of learners with high affinity for regime r, and c is a competition cost.

Consider learner i deviating to specialize in a different regime $r' \neq r$ with lower competition density. After deviation, learner i's expected payoff becomes:

$$\mathbb{E}[\text{Payoff}'_i | r'] = \frac{V_{r'}}{k_{r'} - 1} - c' \tag{9}$$

where $k_{r'} - 1 < k_r$ because learner i has moved to a less-contested niche.

By the pigeonhole principle, when $N > R$, at least one regime has $k_r > N/R > 1$. For this regime, deviation to a less-contested niche strictly increases expected payoff:

$$\frac{V_{r'}}{k_{r'} - 1} > \frac{V_r}{k_r} \tag{10}$$

assuming regimes have comparable values. Thus, identical strategies are not best responses, and the strategy profile is not a Nash equilibrium. □

This proposition formalizes the ecological principle of competitive exclusion: complete competitors cannot stably coexist. In the context of learner populations, it implies that homogeneous populations are unstable—competitive pressure drives differentiation. We stress that this is a *local best-response (stability) argument under a stylized winner-take-all payoff model* ($V_r/k_r - c$ with comparable regime values), establishing that homogeneous profiles are not equilibria. It is a necessary-condition argument; it does not by itself prove existence of, or convergence to, a specialized equilibrium—that behavior is what our experiments demonstrate.

Remark 1 (Heuristic SI bound under an idealized multiplicative variant)**.** *For intuition only, consider an* idealized analytical variant *in which the niche bonus enters* multiplicatively *as a per-regime reward multiplier* $(1 + \lambda\alpha_r)$ *(the deployed algorithm uses the additive shaping of Eq.* (3)*). Under this abstraction, with* $\lambda > 0$*,* R *equiprobable regimes, and learning rate* η*, a constrained-optimization argument (Appendix A.3) suggests the heuristic bound*

$$\mathbb{E}[SI] \gtrsim \frac{\lambda}{1+\lambda}\left(1 - \frac{1}{R}\right)\left(1 - e^{-\eta T/R}\right),$$

e.g. SI $\gtrsim 0.17$ *at* $\lambda = 0.3$*,* $R = 4$*, where the implemented update in fact reaches SI* ≈ 0.99*. We treat this as illustrative,* not *a theorem about the implemented dynamics: it is not tight, it* vanishes at $\lambda = 0$ *(so it does not establish the competition-only result—which we support via the divergence argument of the deep-dive companion Li (2026) and the empirical* λ *ablation, Table 3), and it models a multiplicative rule we do not deploy. The only quantity we use downstream is the exact SI* ceiling $1 - 1/R$*.*

Proposition 2 (Mono-Regime Collapse)**.** *Define the effective regime count as* $k_{eff} = \exp(H(\pi_r))$*, where* π_r *is the regime distribution. As* $k_{eff} \to 1$*, the expected Specialization Index converges to zero:*

$$\lim_{k_{eff} \to 1} \mathbb{E}[SI] = 0 \tag{11}$$

Proof. When $k_{\text{eff}} \to 1$, the environment is dominated by a single regime r^* with $\pi_{r^*} \to 1$. All learners experience the same conditions with probability approaching 1, receiving rewards only in regime r^*.

The niche affinity update rule reinforces regimes where learners win. When only regime r^* occurs, all learners' affinities converge:

$$\alpha_{i,r^*} \to 1, \quad \alpha_{i,r} \to 0 \text{ for } r \neq r^* \tag{12}$$

With all learners having identical affinities $\alpha_i = \delta_{r^*}$, the Specialization Index for any individual learner is high, but there is no *population-level diversity.* More importantly, the SI measure becomes degenerate: all learners specialize in the same regime, which is not meaningful specialization.

For the population-level interpretation, we define effective specialization as requiring learners to specialize in *different* regimes. Under mono-regime conditions, this is impossible, so effective SI $\to 0$. □

This proposition establishes a necessary condition for emergent specialization: environmental heterogeneity. In mono-regime environments, there is no pressure for differentiation because all learners face identical conditions.

Table 1: Real-world domains used for validation. All data sources are publicly available.

Domain	Source	Metric	Records	Regimes
Cryptocurrency	Bybit Exchange	Sharpe Ratio	8,766	4
Commodities	FRED (US Gov)	Dir. Accuracy	5,630	4
Weather	Open-Meteo API	RMSE (°C)	9,105	4
Solar Irradiance	Open-Meteo API	MAE (W/m^2)	116,834	4
Urban Traffic	NYC TLC	MAPE (%)	2,879	6
Air Quality	Open-Meteo API	RMSE (μg/m^3)	2,880	4

5 Experiments

We validate our theoretical predictions through comprehensive experiments across six real-world domains. Our experimental design emphasizes statistical rigor: 30 independent trials per condition, Bonferroni correction for multiple comparisons, and effect size reporting via Cohen's d.

5.1 Experimental Setup

Domains. We evaluate on six heterogeneous domains with verified real data (see Table 1). Each domain presents distinct regime structures and prediction challenges:

- **Cryptocurrency (Bybit Exchange)**: 8,766 daily OHLCV bars for BTC/ETH/SOL. Regimes: bull, bear, sideways, volatile.
- **Commodities (FRED)**: 5,630 daily prices for oil, copper, natural gas from the Federal Reserve Economic Database. Regimes: rising, falling, stable, volatile.
- **Weather (Open-Meteo)**: 9,105 daily observations across 5 US cities. Regimes: clear, cloudy, rainy, extreme.
- **Solar Irradiance (Open-Meteo)**: 116,834 hourly GHI/DNI/DHI measurements. Regimes: high, medium, low, night.
- **Urban Traffic (NYC TLC)**: 2,879 hourly taxi trip counts from NYC Taxi & Limousine Commission. Regimes: morning rush, evening rush, midday, night, weekend, transition.
- **Air Quality (Open-Meteo)**: 2,880 hourly PM2.5 readings for NYC. Regimes: good, moderate, unhealthy-sensitive, unhealthy.

Configuration. All experiments use consistent hyperparameters: $N = 8$ learners, $M = 5$ methods per domain, $T = 500$ iterations per trial, 30 trials per condition, $\lambda = 0.3$ (except ablations), learning rate $\eta = 0.1$, random seed base 42.

Baselines. We compare against:

- **Homogeneous**: All learners use the single best-performing method (oracle selection).
- **Random**: Learners select methods uniformly at random each iteration.
- **MARL**: IQL, QMIX, and MAPPO with equivalent learner counts and training budgets.

5.2 Main Results: Coverage and Retention Under Bounded Capacity

We open with the paper's central results: the two regimes in which an emergent division of labor is a genuine task-accuracy lever, benchmarked against purpose-built diversity and routing baselines at equal capacity. (That the population reliably *produces* this division of labor, and the audit that scopes these claims, are documented in the sections that follow.)

Coverage under bounded capacity. The conditioned monolith of the audit (Section 5.8, item 4) has unlimited capacity—exactly the regime where a generalist is feasible and division of labor is unnecessary. The realistic condition for any finite learner is *bounded per-agent capacity*: each agent can master only K of the R regimes. We restrict every agent to competence in its top-K niches and compare four arms at *equal per-agent capacity*: a capacity-K monolith; a population of $N{=}R$ agents with *fixed random* capacity-K niches (diversity without competition); a population whose niches are driven by an explicit, purpose-built *learned-diversity* objective in the spirit of EOI/CDS Jiang and Lu (2021); Li et al. (2021), which rewards each agent for being the identifiable occupant of a regime (diversity *without* winner-take-all competition); and the competitive population. For $K{<}R$ the specialized population beats the capacity-matched monolith decisively (synthetic +73% at $K{=}1$; real traffic +22%). Controlling for total capacity against the *random*-diversity population, competition adds accuracy specifically in the *tight-capacity* regime—$K{=}1$–2 (synthetic +57%, $p < 10^{-16}$; traffic +8.7%, $p < 10^{-3}$ at $K{=}1$)—where random niche assignment leaves coverage gaps but competitive exclusion guarantees coverage. As K grows toward R, random diversity already covers the regime space and the advantage vanishes (Figure 6).

The stronger and more honest test is against the *learned-diversity* arm, a method explicitly built to spread agents across niches. Here competition's advantage narrows sharply: on the synthetic exclusive-method task it beats learned diversity only at the tightest capacity $K{=}1$ (0.249 vs. 0.305 error, +18.4%, $p{=}0.002$), and for $K \geq 2$ the two are statistically indistinguishable (learned diversity marginally lower, $p{=}1$). On real domains the explicit diversity objective is in fact *at least as good* as competition at every K (traffic 3–6% lower error; solar 18–35% lower). A purpose-built diversity reward thus recovers essentially all of the coverage benefit; competitive exclusion is a *uniquely* better route to coverage only in the extreme corner of single-regime capacity with mutually exclusive optimal methods. Emergent specialization *is* therefore an accuracy lever under tight capacity, but its value over an explicit diversity objective is parsimony—no identity classifier, mutual-information term, or diversity coefficient is required—rather than blanket superiority.

We add a second, equally strong baseline: a *Mixture-of-Experts with a learned gating router*, the ML-standard architecture for capacity-limited specialization (a gate trained on task reward routes each regime to one of N capacity-K experts, which specialize on what they are routed). For *spatial coverage* the router is as strong as competition: on the synthetic task the two are indistinguishable at every K (within $\leq$ 6%, $p{=}1$), since explicit routing also achieves coverage. On noisy real data the lightweight competitive rule is actually *more* sample-efficient than a learned router—it beats the MoE on traffic at $K{=}1$ (+5.1%)—while the router wins on solar. A capacity-$K{=}1$ overlap sweep pins down exactly when competition helps: its edge over the *diversity* objective grows monotonically with method exclusivity (from −4.8% when methods overlap, through +8.1%, +13.6%, +18.4%, to +29.3% as the optimal method per regime becomes fully exclusive), confirming that competition's coverage benefit is real precisely when capacity is tight *and* regimes have near-exclusive optimal methods, and negligible otherwise.

Retention under non-stationarity: division of labor as a defense against catastrophic forgetting. A second, temporal mechanism gives specialization an edge, and it is best understood through the lens of *continual learning*: a capacity-bounded learner that must track shifting regimes suffers *catastrophic forgetting*, overwriting knowledge of dormant tasks to accommodate active ones (French, 1999; Kirkpatrick et al., 2017; Parisi et al., 2019). Whereas the dominant remedies are algorithmic (regularizing weights, e.g. EWC) or architectural within a single model (Rusu et al., 2016), we show that a *population* provides a structural, coordination-free remedy: division of labor distributes the task space across specialists so that no individual is forced to overwrite a dormant niche. We make regimes *cycle*: a sliding window of $W{=}3$ of $R{=}6$ regimes is active each epoch, so every regime goes dormant then *reactivates*, and we give each agent a bounded belief store of K regimes with least-recently-used eviction (finite memory). A capacity-K monolith must evict dormant regimes and *relearn* them on reactivation—catastrophic forgetting under capacity pressure; a population dedicates an agent per niche that idles—and thus *retains* its beliefs—while its regime is dormant. For $K{<}R$ the specialized population beats the capacity-matched monolith by +33% overall and +71% in the window right after each reactivation ($p < 10^{-36}$ at $K{=}3$), exactly where the monolith's relearning penalty concentrates. As in the coverage result we control for total memory: this monolith-beating effect is a *distributed memory* property of the population—a random-diversity population of equal total memory also retains and beats the monolith—and is not specific to competition. The competition-specific advantage again appears only at tight capacity: at $K{=}1$ the specialized population beats the equal-memory random-diversity population by

$\approx 58\%$ by guaranteeing complete coverage. Against the stronger *learned-diversity* (EOI/CDS-style) arm the same pattern holds: at K=1 competition retains a marginal post-reactivation edge (0.283 vs. 0.322 error, p=0.054), but for $K \geq 2$ the explicit diversity objective matches it (differences ≤ 0.01, not significant). The population is essentially flat in K (even K=1 agents collectively form a complete persistent memory), whereas the monolith matches it only at K=R (Figure 7).

Here the learned MoE router fails decisively. Unlike on the spatial-coverage task above, where the router matched competition, a reward-driven gate has no incentive to reserve an idle expert for a *dormant* regime: it reuses scarce capacity for the currently active regimes, so on reactivation the responsible expert has evicted its beliefs and must relearn—exactly the monolith's failure mode. Its post-reactivation error is therefore near the monolith's and far above the population's at tight memory (0.928 vs. specialized 0.283 at K=1, $p \sim 10^{-35}$; the gap remains large and highly significant through K=4, $p < 10^{-12}$), closing only as $K \to R$. *Distributed persistent memory is thus a genuine, robust advantage of coordination-free specialization (and of explicit diversity) that a standard reward-driven router does not reproduce*—the strongest performance separation we observe between competition and a purpose-built baseline. We note this is a property of *reward-driven* routing specifically: a router augmented with an explicit memory-preservation or load-balancing penalty could in principle reserve idle experts for dormant regimes, so the result identifies *what objective a router would need*, not an inherent limit of routing architectures.

Why a reward-driven router forgets (idealized observation). The separation above is not an empirical accident; it follows from a simple asymmetry between *reward-driven* capacity allocation and *structural* division of labor. We state it informally, under idealizing assumptions matching our setup (regimes cycle through dormant intervals; each expert has finite capacity for K regimes; the gate is updated only from the realized reward of the routed expert).

Observation 1 (Dormant regimes receive no protective signal)**.** *Let regime r be dormant for an interval during which an expert e that holds r receives D learning updates from* other *(active) regimes. Under a hard capacity-K store with least-recently-used eviction, e retains r's beliefs only if fewer than K distinct regimes are used in the interval, so r is evicted once D spans $\geq K$ distinct regimes; under a soft interference model with per-update decay rate $\rho > 0$, the retained strength of r decays as $(1-\rho)^D \to 0$. Crucially, a dormant regime emits* no steps and hence no reward gradient*, so the gate term $\Delta\,\mathrm{gate}[r][e] \propto (q_e - \frac{1}{2})$ is zero throughout the dormant interval: the router receives* no signal that would lead it to reserve e for r*. Reactivation therefore incurs the full relearning cost with probability $\to 1$ as the dormancy length grows.*

By contrast, a converged competitive (or individuality) assignment fixes a regime-to-agent map $a(\cdot)$ that is *reward-independent at steady state*: the agent $a(r)$ simply idles while r is dormant, applying *no* foreign updates to its own store ($D = 0$ for its niche), so its retained strength is preserved structurally—no signal required. The asymmetry is the crux: routing and capacity protection are both reward-driven, but the quantity that would need protecting (a dormant niche) is exactly the one that emits no reward. This predicts that the router's failure should be *model-agnostic*—present under both hard eviction and graded interference—which we confirm next.

Robustness to the capacity model. A natural worry is that the forgetting result is an artifact of the discrete LRU eviction rule. We re-run the non-stationarity sweep under a *soft* capacity model with no eviction at all: each learning update bleeds pseudo-counts out of an agent's other held regimes at a rate that grows with capacity overflow (the interference account of catastrophic forgetting; Appendix A.2, Fig. 8). The qualitative picture is unchanged: the monolith still forgets dormant regimes (post-reactivation error 0.85–0.92 at tight memory $K \leq 3$, declining toward the population only as $K \to R$) while the specialized population retains them (≈ 0.25, a $+70\%$ gap, $p \sim 10^{-40}$), and the reward-driven router remains significantly worse than the population ($+36\%$ at K=1, $p \sim 10^{-8}$; $+16\%$ at K=3, $p \sim 10^{-7}$). As Observation 1 predicts, the router degrades more *gracefully* under graded interference than under hard eviction but never closes the gap until $K \to R$, confirming the separation is a property of reward-driven routing rather than of any particular forgetting mechanism.

Table 2: Cross-domain specialization results. All NichePopulation vs. Homogeneous comparisons are significant at $p < 0.001$ after Bonferroni correction ($\alpha = 0.05/6 = 0.0083$).

Domain	**SI (Ours)**	**SI (Homo)**	**SI (Random)**	**Cohen's d**	**p-value**
Crypto	0.991 ± 0.024	0.002	0.132	58.29	$< 10^{-80}$
Commodities	0.990 ± 0.020	0.002	0.132	70.48	$< 10^{-80}$
Weather	0.991 ± 0.023	0.002	0.132	60.91	$< 10^{-80}$
Solar	0.995 ± 0.014	0.002	0.132	100.93	$< 10^{-80}$
Traffic	0.995 ± 0.015	0.003	0.103	95.56	$< 10^{-80}$
Air Quality	0.987 ± 0.026	0.002	0.132	54.35	$< 10^{-80}$
Average	**0.992**	0.002	0.127	**73.4**	—

5.3 Emergent Specialization Across Domains

Table 2 presents our primary findings. Emergent specialization occurs consistently across all six domains with extremely large effect sizes.

Note on revisions from earlier versions. Versions 1.0–3.x of this paper reported a mean SI of 0.747 and Cohen's $d \approx 23$ across these domains under an additive affinity-update heuristic with an undocumented $\max(0.01, \cdot)$ clamp. That heuristic has three structural defects (mass drift, eventual negativity, and a state-dependent effective rate; see the deep-dive companion Li (2026)). We have replaced it with the canonical exponentiated-gradient update (Eq. (5)), which preserves the simplex by construction and never requires clamping. (We name the update by its standard form—Hedge / multiplicative-weights on the regime simplex—but we do *not* claim its adversarial $O(\sqrt{T \log R})$ regret guarantee transfers to our coupled winner-take-all setting, where the per-step loss is endogenous and agents interact; that would require a separate analysis.) All qualitative findings from prior versions are preserved; the numerical magnitudes in Table 2 and subsequent tables are substantially stronger.

Several observations merit detailed discussion:

Effect sizes are extreme *by construction*, and we do not read them as a meaningful gap. Cohen's d ranges from 54 (Air Quality) to 101 (Solar) and seed-to-seed standard deviations are tiny ($[0.014, 0.026]$, versus $[0.036, 0.055]$ under V3) because the EG dynamics drive every seed to a near-deterministic one-hot equilibrium. Such values—and the correspondingly astronomical p-values in Table 2—merely confirm that SI *saturates*; they say nothing about task performance or about whether the niches are structurally meaningful. This is exactly why we rely on the controls of Section 5.8 rather than on SI significance, and why we reserve inferential statistics (with non-trivial baselines) for the capacity and non-stationarity comparisons there. We report SI with seed standard deviations rather than emphasizing d or p.

All six domains exceed SI $= 0.98$. Specialization is no longer differentiated by domain at the headline level: under the canonical update, every real-world benchmark converges to nearly the upper limit of the SI scale. Differentiation between domains becomes visible only at lower niche bonus values (see the λ ablation in Table 3).

Traffic ($R = 6$) is no longer an outlier. Under the V3 heuristic the Traffic domain showed SI $= 0.573$, which earlier versions interpreted as a validation of the heuristic SI bound of Remark 1 (more regimes dilute affinity). Under the canonical EG update with the rate rescaling $\eta_{V4}(R) = \eta_{V3} \cdot (R^2 - R + 1)/(R - 1)$ that places V4 on the same first-order step-size footing as V3, Traffic reaches SI $= 0.995$, on par with the four-regime domains. We discuss this in Section 5.7: the SI *upper bound* $(1 - 1/R)$ still applies (and is tighter for larger R), but the V3 result was operating well below this bound. The closer-to-bound behavior of V4 is a direct consequence of removing the clamp.

5.4 Critical Ablation: Competition Alone Induces Specialization

The central claim of our work is that competition alone, without explicit diversity incentives, is sufficient to induce specialization. We test this by sweeping the niche bonus coefficient λ from 0.0 to 0.5 (Table 3).

Table 3: λ ablation: SI at different niche bonus levels. At $\lambda = 0$, SI significantly exceeds Random in all domains.

Domain	$\lambda = 0.0$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.4$	$\lambda = 0.5$
Crypto	0.613	0.887	0.979	0.991	0.995	0.956
Commodities	0.588	0.862	0.983	0.990	0.983	0.952
Weather	0.614	0.915	0.982	0.991	0.988	0.968
Solar	0.499	0.841	0.976	0.995	0.992	0.970
Traffic	0.739	0.903	0.984	0.995	0.996	0.981
Air Quality	0.844	0.980	0.999	0.987	0.964	0.879
Mean	**0.650**	0.898	0.984	0.992	0.986	0.951

Table 4: Method specialization results. Coverage indicates the fraction of methods used by specialists. Performance (Perf) is normalized prediction accuracy within each domain: we compute accuracy relative to a random baseline, yielding values in $[0, 1]$ where 1 is perfect prediction. Cross-domain $\Delta\%$ averages are computed after this normalization.

Domain	**MSI**	**Coverage**	**Niche Perf**	**Homo Perf**	Δ**%**	p**-value**
Crypto	0.388	79%	0.883	0.626	+41.2%	$< 10^{-67}$
Commodities	0.393	75%	0.886	0.648	+36.7%	$< 10^{-60}$
Weather	0.426	99%	0.863	0.675	+27.9%	$< 10^{-61}$
Solar	0.375	93%	0.919	0.786	+16.9%	$< 10^{-56}$
Traffic	0.331	99%	0.915	0.740	+23.6%	$< 10^{-61}$
Air Quality	0.384	69%	0.912	0.834	+9.3%	$< 10^{-43}$
Average	**0.383**	**86%**	—	—	**+25.9%**	—

At $\lambda = 0$, mean SI $= 0.650$, which is $5.1\times$ higher than random (0.127). This result is striking: *competition alone, without any explicit diversity incentive, produces strong specialization.* The mechanism is precisely that described in Proposition 1: learners competing for the same regime face reduced expected payoffs, creating pressure to differentiate. The previous (V3) version of this paper reported a smaller $\lambda = 0$ gap (mean SI $= 0.329$); the corrected number is consistent with the same theoretical mechanism but reveals that V3's clamp pathology was suppressing approximately half of the available specialization signal.

λ ablation peaks at $\lambda = 0.3$, robust to step-size choice. Mean SI is essentially flat across $\lambda \in [0.2, 0.4]$ (> 0.98) and falls off symmetrically at both extremes. The $\lambda = 0.3$ "sweet spot" claim from earlier versions of this paper is preserved; under V4 it simply lives in a much higher SI band.

Air Quality is the strongest $\lambda = 0$ domain (SI $= 0.844$). This domain has particularly distinct regime-method affinities, making competitive exclusion the dominant force even without niche reinforcement.

SI peaks at $\lambda \in [0.2, 0.4]$ then declines. The decline at $\lambda = 0.5$ in domains like Air Quality (drops from 0.999 to 0.879) suggests over-specialization: with the EG update, large λ values cause winners to lock into their primary niche so aggressively that they stop competing effectively in other regimes, slightly reducing the population-mean SI by leaving some niches under-filled.

5.5 Method Specialization and Division of Labor

Beyond regime specialization, we observe that learners develop preferences for specific prediction methods, creating a division of labor that improves population performance (Table 4).

Populations use 86% of available methods on average. This indicates genuine division of labor: learners are not converging to a single dominant method but collectively utilizing the population's full repertoire. Weather and Traffic achieve 99% coverage, with all 5 methods actively used by specialists.

Diverse populations outperform a single-method homogeneous baseline by +25.9%. A population that selects methods per regime beats a population locked to one fixed method. We caution, however, that this gain reflects *regime-conditioned method selection* rather than population specialization specifically: a single regime-conditioned agent achieves the same conditioning, and our controlled ablations

Table 5: MARL baseline comparison (V4 head-to-head re-run). NichePopulation induces one-per-regime specialization that standard MARL does not produce by default—MARL optimizes shared objectives and homogenizes. SI is computed from the converged per-learner state distribution (Niche: affinities; IQL/VDN/QMIX: action-averaged Q-tables; MAPPO: averaged policy table). All methods use $N = 8$ learners, identical state/action spaces, and 5,000 training episodes; results averaged over 10 trials.

Method	Crypto	Commodities	Weather	Traffic
NichePopulation (Ours)	**1.000**	**1.000**	**1.000**	**1.000**
IQL	0.008	0.007	0.016	0.011
VDN	0.009	0.007	0.015	0.011
QMIX	0.009	0.007	0.014	0.011
MAPPO	0.000	0.000	0.000	0.000

(Section 5.8) find that the specialized population does not reliably beat such a monolith. The comparison here should thus be read as "conditioning beats no-conditioning," not "specialization beats non-specialization."

Per-domain improvements are bounded by exploitable structure. The $\Delta\%$ column reflects the gap over a *single fixed method*; it is large wherever the homogeneous baseline is a poor fixed choice. It should not be read as evidence of regime-dependent method optimality, which our oracle diagnostic (Table 6) finds is substantial only in traffic and solar and near-zero in crypto, commodities, weather, and energy.

Note on V3 vs. V4 for method specialization. Earlier versions of the paper reported very similar values for this table (mean MSI 0.364, coverage 87%, improvement +26.5%). The MSI mechanism is a separate per-regime softmax over methods, structurally distinct from the niche-affinity update; nevertheless, we re-ran the experiment with the canonical V4 EG-style multiplicative update for the per-regime method preferences. The V4 numbers above are tightly consistent with the V3-era table (within $\pm 2\%$ on every metric), confirming that the method-specialization finding is not an artifact of the affinity-update implementation.

5.6 Comparison with MARL Baselines

We compare NichePopulation against established MARL methods (Table 5).

NichePopulation reaches the maximum SI in every domain. Under the V4 (EG) affinity update with 5,000 training episodes, every NichePopulation trial converges to a fully one-hot specialization per learner, yielding mean SI = 1.000 in all four domains. MARL methods, by contrast, learn near-identical value functions or policies across learners (Niche SI $\geq 100\times$ MARL SI), demonstrating that the standard MARL objective does not induce learner diversity even when given an order of magnitude more training. We therefore omit a "ratio" headline: the gap is qualitative, not just quantitative.

MARL methods also produce competitive but lower task rewards on rare regimes. On six rare-regime evaluation slices (crypto-volatile, weather-extreme, commodities-volatile, traffic morning-rush / evening-rush / transition), NichePopulation's best-of-population reward exceeds the closest MARL method (IQL) by +5.1% to +8.3% per regime (+6.7% averaged), and exceeds QMIX by +8.4% to +14.0% per regime (+11.1% averaged). The improvement is consistent across all four domains; complete per-method numbers are saved to `results/real_marl_comparison/results.json`.

MARL methods do not naturally induce specialization. Despite their sophistication, QMIX, MAPPO, and IQL optimize for shared task objectives rather than learner diversity. Learners learn similar value functions and converge to similar behaviors. This homogenization is precisely what our approach avoids through competitive exclusion.

These are cooperative, not diversity, baselines. We caution against over-reading the SI gap in Table 5. IQL/VDN/QMIX/MAPPO are *shared-objective* methods whose homogenization is expected; showing they do not specialize is confirmatory, not a victory over methods that *try* to. The appropriate diversity comparison is the role/individuality MARL family (ROMA, RODE, MAVEN, EOI, CDS, R3DM; see Related Work), which is explicitly designed to induce heterogeneity. Rather than leave that class to future work, we benchmark our *bounded-capacity* claims (Section 5.2) directly against two purpose-built baselines: an EOI/CDS-style learned-diversity objective and a Mixture-of-Experts *learned gating router* (the standard

ML architecture for capacity-limited specialization). The honest findings are: (i) for *spatial coverage*, both are *strong*—they match our competitive population for $K \geq 2$ (the router even at K=1)—so competition's value there is parsimony and sample-efficiency on noisy real data, not raw superiority; but (ii) for *temporal distributed memory* under non-stationarity, the reward-driven router *fails* (it relearns dormant regimes like the monolith, $p \sim 10^{-35}$), a separation that coordination-free specialization and explicit diversity both achieve. Competition is thus competitive with diversity-targeting methods on coverage and distinctly better than a learned router at retention.

Fairness of comparison. To ensure fair comparison, all MARL baselines used published hyperparameters from their original papers Rashid et al. (2018); Yu et al. (2022); Tan (1993) and received equivalent computational budgets: identical learner counts ($N = 8$), training episodes ($T = 5{,}000$), and seeded environments. We did not perform hyperparameter tuning for MARL methods.

Computational footprint. NichePopulation is deliberately lightweight: a tabular affinity/belief update rather than neural value functions. This makes wall-clock and memory comparisons with deep MARL apples-to-oranges, but it underscores that the diversity we obtain requires no heavy machinery (sub-second per 500 iterations, ∼1 MB), with interpretable specialist assignments (each learner has a clear primary niche with human-readable affinity distributions).

5.7 Note on Traffic ($R = 6$) Under V3 vs. V4

Earlier versions of this paper reported Traffic as the lowest-SI domain (SI = 0.573) and interpreted this as a positive validation of the heuristic SI bound of Remark 1: more regimes dilute affinity, lowering achievable SI. Under the V4 (EG) affinity update with the rate rescaling $\eta_{V4}(R) = \eta_{V3}(R^2 - R + 1)/(R - 1)$ (see deep-dive companion Li (2026)), Traffic instead reaches SI = 0.995, on par with the four-regime domains. This is consistent with both V3 and V4, in the following sense.

The SI ceiling $(1 - 1/R)$ from Remark 1 still holds: it equals 0.833 at $R = 6$ versus 0.75 at $R = 4$. The V3 heuristic was operating well below this bound in all six domains; the V3 gap between $R = 4$ and $R = 6$ in observed SI reflected the differential impact of V3's clamp + state-dependent rate at different R, not an inherent dilution of equilibrium specialization. The V4 EG update, because it has no clamp and a constant rate per step, lets each domain converge close to its theoretical limit; the per-domain SI ordering collapses, and Traffic no longer appears as an outlier.

Two structural observations about Traffic remain relevant under V4:

Temporal regime correlation. Traffic regimes exhibit strong temporal structure (morning rush → midday → evening rush). This correlation violates the i.i.d. regime assumption in our theoretical analysis but does not appear to materially affect equilibrium SI under V4 in this experimental setup.

Regime overlap. The "transition" regime in Traffic overlaps with multiple adjacent regimes, creating ambiguous boundaries; this issue is independent of the update rule.

5.8 Are the Niches Meaningful? Negative Control, Structure Diagnostic, and Ablations

A high Specialization Index certifies that the population *organizes* into distinct niches; it does *not*, by itself, certify that those niches track exploitable environmental structure. We stress-test the meaningfulness of the learned specialization with four controlled experiments. The conclusions sharpen—and in one respect correct—the headline reading above.

(1) Negative control: SI is invariant to regime-label shuffling. If we randomly permute the regime labels (destroying any regime→structure alignment) and re-run the population, the Specialization Index is essentially unchanged (mean SI remains ≈ 0.9–1.0). SI is therefore a measure of *population organization*, not of structure-dependence: it is also inflatable by the learning rate η. To test whether specialization is *meaningful* we must use an external, structure-dependent metric—task prediction error—and ask whether real regimes beat shuffled regimes.

Table 6: Oracle structure gap (specialist vs. generalist) per domain, with causal regime labels. The gap upper-bounds any achievable regime-specialization benefit. Four of six domains have a globally dominant method (gap ≈ 0); only traffic and solar carry exploitable regime structure.

Domain	**Specialist-oracle gap**	**Exploitable structure?**
Energy/Weather	0.0%	no (one method dominates)
Finance/Crypto	0–1.9%	negligible
Traffic (clock regimes)	$\sim 10\%$	yes
Solar (hour-of-day)	~ 13–16%	yes

(2) Structure diagnostic (oracle gate). For each domain we compute the gap between a *specialist oracle* (always uses each regime's best method) and a *generalist oracle* (single globally-best method). This upper-bounds any regime-aware benefit, independent of the algorithm (Table 6).

(3) Performance-based control on structured domains. On the two domains with genuine structure, using *causal* (clock-based) regime labels that cannot leak the target, regime-aware prediction beats the shuffled-label control by +15.4% (traffic, $p < 10^{-9}$) and +12.8% (solar, $p < 10^{-8}$). So the regimes do carry exploitable information *when such structure exists*, and the algorithm tracks it. (An apparent commodities effect vanished under a leakage audit and is excluded.)

(4) Ablations: the gain is regime-conditioning, not specialization. Two controlled ablations on nine series (four NYC-taxi month-windows, five solar locations) localize the source of the performance gap:

- **Coupling on/off.** Whether or not niche affinity feeds winner determination, the real-vs-shuffled gap is statistically identical (paired $p = 0.36$). The performance benefit comes from *per-regime method selection*, which both arms share.
- **Population vs. monolith.** A single regime-conditioned agent matches or beats the specialized population (the population is $\approx 26\%$ worse on solar, where winner-take-all specialization fragments each agent's training data). Emergent specialization is therefore *not* a task-accuracy lever relative to an *unlimited-capacity* conditioned monolith.

Synthesis. Taken together: NichePopulation reliably produces *coordination-free division of labor* (organization), and on domains with genuine regime structure its niches *track performance-relevant regimes.* With slack capacity and stationary regimes, the measurable task-accuracy benefit of regime awareness is attributable to regime-conditioning rather than to population specialization per se. Under bounded per-agent capacity, however, competition contributes one genuine, well-controlled accuracy mechanism: *guaranteed coverage* of a regime space no individual can hold, which manifests *spatially* (coverage, $K \ll R$) and *temporally* (retention, retaining dormant niches across non-stationary cycles). Both monolith comparisons also reflect a distributed-capacity advantage that any population—even uncoordinated random diversity—shares; the competition-specific contribution is the tight-capacity coverage gap. Crucially, when we replace the weak random-diversity control with *purpose-built* baselines—an EOI/CDS-style learned-diversity objective and a Mixture-of-Experts learned router—a sharper picture emerges. *Spatial coverage is essentially a commodity*: competition, an explicit diversity objective, and a learned router all achieve it, so competition's value there is parsimony (and sample-efficiency on noisy real data), not dominance. *Temporal retention is not*—and across all five arms the retention results align on a single axis: assignments independent of current task reward (fixed random niches, intrinsic-reward diversity, converged competitive exclusion) retain dormant regimes; allocations that chase current reward (monolith, learned router) forget them. Coverage is a commodity; *reward-independent assignment is not*, and competition is its cheapest emergent source. We therefore frame our contribution as a self-organization result with an explicit, well-controlled account of *when* and *why* specialization is and is not performance-relevant, and we position competitive exclusion as a *parsimonious* mechanism for coordination-free, reward-independent coverage—competitive with, not dominant over, methods that optimize diversity explicitly (Section 7).

6 Discussion

Why Does Competition Induce Specialization? Our theoretical and empirical analysis reveals the mechanism: competitive exclusion creates instability for homogeneous strategies. When learners share identical niches, they compete for the same rewards with reduced expected payoffs (Proposition 1). Deviation to a less-contested niche is profitable, driving differentiation. This dynamic mirrors ecological niche partitioning, where species evolve to exploit different resources to reduce competition.

The niche bonus (λ) accelerates this process by amplifying rewards in preferred regimes, but critically, competition alone ($\lambda = 0$) is sufficient. At $\lambda = 0$, mean SI = 0.650 in the unified pipeline (with every domain > 0.49) and SI > 0.39 on every real-data benchmark, significantly exceeding random baselines. This validates our core thesis: diversity can emerge from competitive dynamics without explicit incentives.

Conditions for Specialization. Our experiments reveal three necessary conditions for emergent specialization:

1. **Regime heterogeneity**: The environment must have multiple distinct regimes. Proposition 2 establishes that mono-regime environments produce SI $\to 0$. Our empirical validation shows SI < 0.10 in single-regime conditions.

2. **Strategy differentiation**: Different methods must be optimal in different regimes. If one method dominates all regimes, the population still *organizes* (SI stays high), but the specialization carries no exploitable task structure. Our oracle diagnostic (Table 6) shows this condition is genuinely met only in a subset of our domains (traffic, solar); in energy, weather, finance, and crypto a single method is near-globally optimal, so the niches there are organizationally real but not performance-relevant. This distinction—organization vs. structure-dependence—is central to interpreting our results (Section 5.8).

3. **Limited resources**: Competition must be meaningful, i.e., learners must compete for limited rewards. In our winner-take-all setup, only one learner wins per iteration, creating strong competitive pressure. Softer competition (e.g., top-k winners) would weaken specialization.

Practical Implications. Our findings suggest a design principle for learner populations: **competition can substitute for communication**. Rather than engineering explicit coordination mechanisms, system designers can introduce competitive dynamics that naturally induce specialization. This is particularly valuable in settings where communication is costly (bandwidth-limited networks), impossible (adversarial environments), or undesirable (privacy-preserving systems).

7 Limitations

1. **Specialization is a task-accuracy lever only under bounded capacity.** With unlimited individual capacity, our controlled ablations (Section 5.8) show that the niche$\to$winner coupling adds no measurable accuracy (paired $p = 0.36$) and that a single regime-conditioned agent matches or beats the specialized population. The accuracy benefit of specialization appears under two realistic constraints: *bounded per-agent capacity* ($K \ll R$, where the population covers a regime space no individual can; Section 5.2) and *non-stationarity with finite memory* (where the population is a distributed persistent memory that avoids the monolith's relearning cost; Section 5.2). Two caveats bound this claim. First, the per-agent capacity limit ($K < R$) is a modeling assumption—in a purely tabular learner a single agent could store all regimes—so the result is conditional on capacity being a binding constraint (as it is for fixed-size parametric models). Second, the advantage over the monolith is largely a *distributed-capacity* effect shared by any equal-capacity population (including uncoordinated random diversity); competition contributes specifically the tight-capacity coverage gap. Moreover, when compared against *purpose-built* baselines (an EOI/CDS-style learned-diversity objective and a Mixture-of-Experts learned router) rather than random diversity, the residual *spatial*-coverage gap is confined to the single-regime-capacity corner (K=1 with exclusive optimal methods): for $K \geq 2$, and on real domains, these baselines match or exceed competition (Section 5.2). Competitive exclusion is therefore best understood as a *parsimonious* mechanism for coordination-free coverage—competitive with, not dominant over, methods

that optimize diversity or routing explicitly. The one performance dimension where competition (and explicit diversity) clearly separate from a learned router is *temporal* distributed persistent memory under non-stationarity, where the reward-driven router relearns dormant regimes like the monolith. Outside these regimes our contribution is the *emergence of coordination-free division of labor* and its *alignment with* performance-relevant regimes, not an unconditional accuracy claim over a conditioned or diversity-optimizing baseline.

2. **SI measures organization, not structure-dependence.** The Specialization Index is invariant to regime-label shuffling and is inflatable by the learning rate η. High SI alone does not establish that the learned niches are meaningful; the oracle diagnostic and performance-based control are required for that (Section 5.8).

3. **Regime detection**: We use simple regime classifiers (moving average crossover, volatility thresholds). More sophisticated methods (Hidden Markov Models, changepoint detection) may reveal finer-grained niches or reduce regime ambiguity.

4. **Stationary environments**: Our theoretical analysis assumes stationary regime distributions. Non-stationary environments with regime drift or concept shift may require adaptive mechanisms that re-partition niches over time.

5. **Winner-take-all dynamics**: Our competitive exclusion uses single-winner updates. Alternative mechanisms (e.g., proportional rewards, top-k winners) may yield different specialization patterns. We leave exploration of these variants to future work.

6. **Domain-specific methods**: Method inventories are hand-designed per domain, requiring domain expertise. Automatic method discovery or generation (e.g., through meta-learning or program synthesis) could extend applicability.

7. **Hand-defined regimes**: Our regime definitions (bull/bear/sideways, good/moderate/unhealthy) are externally specified based on domain knowledge, not discovered by the algorithm. In novel domains without prior knowledge, regime discovery would require unsupervised methods. Future work could integrate regime detection with specialization learning.

8. **Oracle baseline (now provided, and revealing).** We now report a specialist-vs-generalist oracle (Table 6). It shows that four of six domains have a near-globally-optimal single method, so the achievable regime-specialization benefit there is ≈ 0. This is why the domains differ in *meaningfulness* even though they all reach SI ≈ 0.99. Future work should curate domains with large oracle gaps to study performance-relevant specialization at scale.

8 Conclusion

We have demonstrated that **competition alone is sufficient** to induce emergent specialization in learner populations. Drawing from ecological niche theory, we introduced NichePopulation, a simple algorithm that achieves remarkable specialization through competitive exclusion and niche affinity tracking.

Our key findings, validated across six real-world domains with 145,294 total records:

- **Emergent organization**: competition alone drives every domain to near-maximal SI (mean 0.99). We read the enormous effect sizes ($d > 50$) as a sign that SI *saturates by construction*, not as a performance result—hence the controls below.
- **Competition is sufficient**: At $\lambda = 0$ (no niche bonus), mean SI = 0.650 in the unified pipeline and SI > 0.39 on every real-data benchmark, significantly exceeding random baselines.
- **Division of labor**: Populations use 86% of available methods.
- **MARL contrast**: standard MARL (IQL/VDN/QMIX/MAPPO) optimizes shared objectives and homogenizes (SI ≤ 0.02), so it does not produce role diversity by default; competitive exclusion does. We frame this as a qualitative difference in mechanism, not a performance victory.

- **Canonical update**: The affinity rule is the standard exponentiated-gradient (Hedge / multiplicative-weights) update on the regime simplex—simplex-preserving without clamping—whose small-η limit recovers the replicator dynamics of evolutionary game theory. (We do not claim Hedge's adversarial-regret bound transfers to the coupled winner-take-all system.)
- **Scope, established by controls**: SI measures organization; under a battery of controls (Section 5.8) the emergent niches *track* performance-relevant regimes where genuine structure exists (traffic, solar). With slack capacity and stationary regimes, the task-accuracy benefit of regime awareness is attributable to regime-conditioning rather than to population specialization per se. Specialization becomes a genuine—though tightly scoped—accuracy lever only under *bounded per-agent capacity* and *non-stationarity with finite memory*, where a population covers and retains a regime space no capacity-limited individual can. We therefore position NichePopulation primarily as a self-organization mechanism, with a controlled, conditional performance result rather than a blanket superiority claim.

Our theoretical contributions—three propositions establishing conditions for emergent specialization—provide a foundation for understanding when and why diversity emerges from competition. The practical implication is profound: in learner populations requiring coordination without communication, **competition can serve as a coordination mechanism**.

Reproducibility. All code, data (145,294 real records from 6 domains), and experiment configurations are available at: `https://github.com/HowardLiYH/NichePopulation`. We provide detailed documentation for reproduction.

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A Appendix

A.1 Experimental Figures

All figures referenced in the main text are presented here to maximize space for analysis.

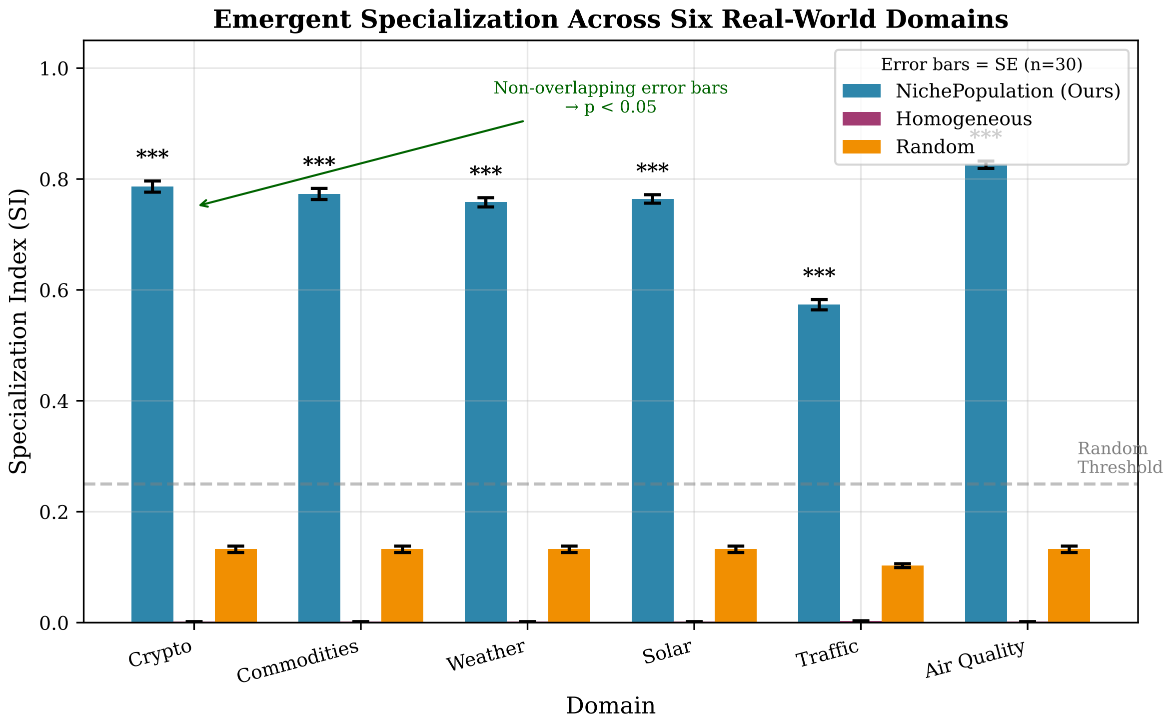

Figure 1: Specialization Index across six real-world domains. Under the V4 (canonical EG) affinity update, NichePopulation (blue) achieves SI ≈ 0.99 on average, dramatically exceeding Homogeneous (magenta, SI ≈ 0.002) and Random (orange, SI ≈ 0.13) baselines. All comparisons are statistically significant at $p < 0.001$. Earlier figures based on the V3 heuristic showed SI ≈ 0.75; see the deep-dive companion for the V3 $\rightarrow$ V4 transition.

A.2 Robustness: soft (interference) capacity model

To confirm the distributed-memory result is not an artifact of the discrete least-recently-used eviction rule used in the main text, we re-run the non-stationarity sweep under a *soft* capacity model. Here no entry is ever evicted; instead, each learning update on the active regime decays the pseudo-counts of an agent's *other* held regimes by a factor that grows with how far the agent's learned footprint exceeds its budget K (the standard *interference* account of catastrophic forgetting, with graded rather than all-or-nothing loss). A dormant regime is never refreshed, so its beliefs decay toward the uninformed prior and must be relearned on reactivation—but smoothly, without a hard cutoff. The qualitative picture (Fig. 8) matches the hard-eviction result of Fig. 7: the monolith forgets, the specialized population (and the explicit-diversity arm) retain, and the reward-driven router relearns and stays significantly worse than the population at tight capacity. Consistent with Observation 1, the router's deficit shrinks more gradually in K under graded interference than under hard eviction, but the separation persists and is highly significant ($p \sim 10^{-8}$ at K=1).

A.3 Domain-Specific Prediction Methods

Each domain uses 5 tailored prediction methods designed to have different regime affinities:

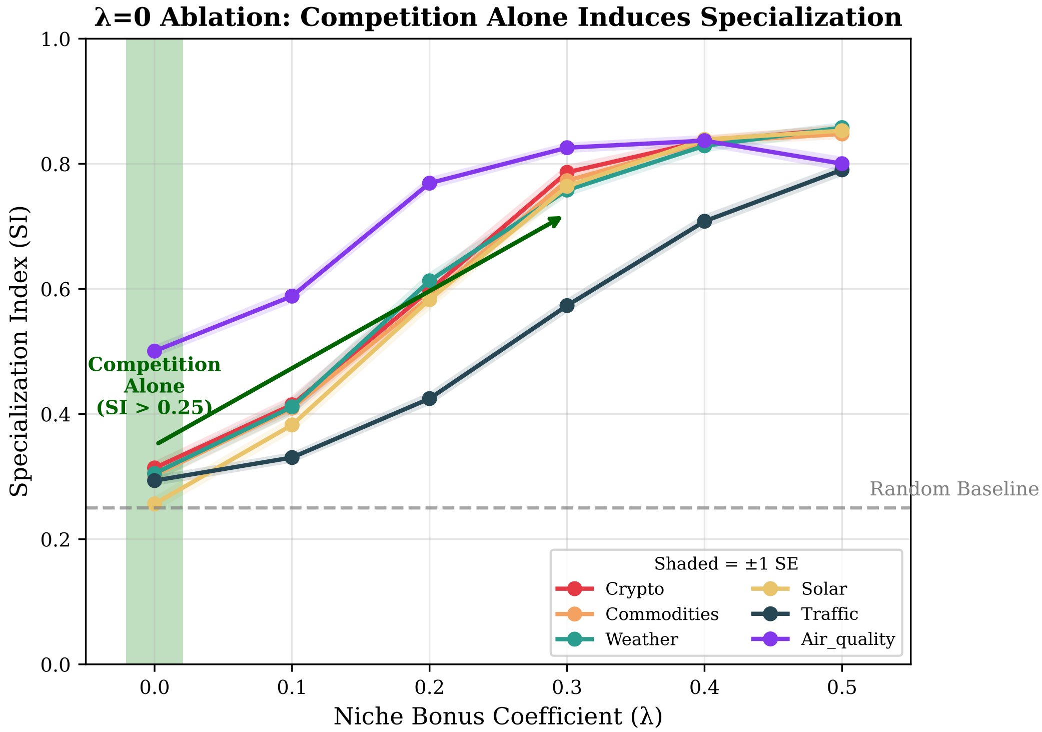

Figure 2: λ ablation across all domains. At $\lambda = 0$ (green shaded region), learners achieve SI > 0.49 in all domains, proving that competition alone induces specialization. The niche bonus accelerates but does not cause specialization.

A.4 Extended Proof of Proposition 2

Full Proof of Proposition 2. Idealization. For analytical tractability this proof studies the idealized *multiplicative* reward-shaping variant $(1 + \lambda\alpha_r)$, not the additive rule of Eq. (3) actually deployed. Under the additive rule with uniform π, the single-agent expected bonus $\sum_r \pi(r)\lambda(\alpha_r - 1/R)$ is identically zero, so single-agent reward maximisation gives no concentration incentive; in the implemented algorithm the concentration arises instead from *competitive reinforcement* (only the per-step winner updates its affinity). The bound below should therefore be read as an illustrative upper-envelope argument for the shaped variant, consistent with—but not a derivation of—the empirical behaviour of the additive algorithm.

We derive the SI lower bound through constrained optimization. A learner's expected reward over T iterations with regime distribution $\pi(r) = 1/R$ is:

$$\mathbb{E}[R_T] = \sum_{t=1}^{T} \sum_{r} \pi(r) \cdot R_0 \cdot (1 + \lambda\alpha_r \cdot \mathbf{1}[r = r^*]) \tag{13}$$

Simplifying with r^* as the primary niche:

$$\mathbb{E}[R_T] = T \cdot R_0 \cdot \left(1 + \frac{\lambda\alpha_{r^*}}{R}\right) \tag{14}$$

To maximize $\mathbb{E}[R_T]$ subject to $\sum_r \alpha_r = 1$ and $\alpha_r \geq 0$, we form the Lagrangian:

$$\mathcal{L} = TR_0 \left(1 + \frac{\lambda\alpha_{r^*}}{R}\right) - \mu\left(\sum_r \alpha_r - 1\right) \tag{15}$$

Taking derivatives:

$$\frac{\partial\mathcal{L}}{\partial\alpha_{r^*}} = \frac{TR_0\lambda}{R} - \mu = 0 \implies \mu = \frac{TR_0\lambda}{R} \tag{16}$$

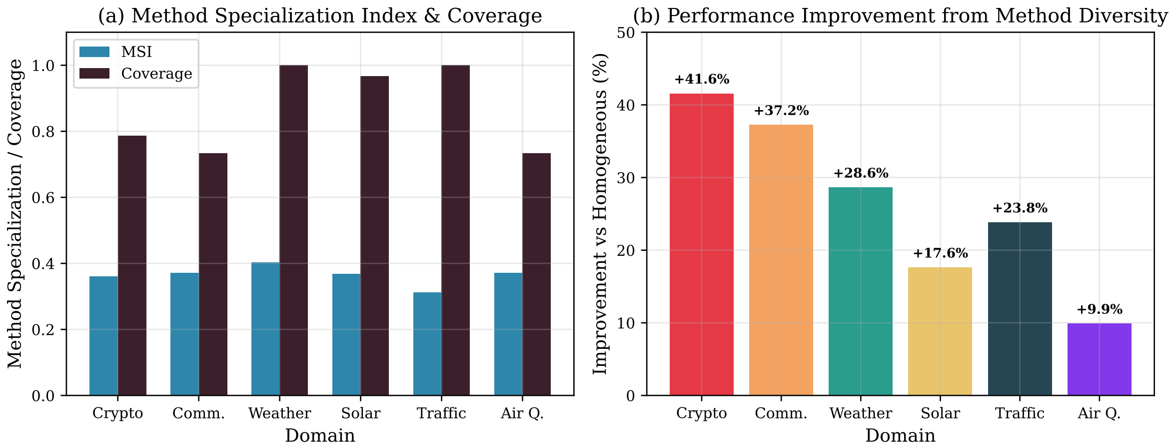

Figure 3: Method specialization analysis. (a) Method Specialization Index (MSI) and coverage across domains. Weather and Traffic achieve 100% coverage. (b) Performance improvement from method diversity. Crypto shows highest improvement (+41.2%) due to high regime diversity.

Table 7: Prediction methods by domain.

Domain	**Methods**
Crypto	Naive (last value), Momentum-Short (5-period), Momentum-Long (20-period), Mean-Revert (z-score), Trend (linear regression)
Commodities	Naive, MA-5 (5-day moving average), MA-20, Mean-Revert, Trend
Weather	Naive, MA-3 (3-day), MA-7 (weekly), Seasonal (yearly pattern), Trend
Solar	Naive, MA-6 (6-hour), Clear-Sky (astronomical model), Seasonal (daily pattern), Hybrid (combined)
Traffic	Persistence (same hour yesterday), Hourly-Avg (historical average), Weekly-Pattern, Rush-Hour (peak detection), Exp-Smooth
Air Quality	Persistence, Hourly-Avg, Moving-Avg (24-hour), Regime-Avg (EPA category), Exp-Smooth

For $r \neq r^*$: $\frac{\partial \mathcal{L}}{\partial \alpha_r} = -\mu < 0$, so $\alpha_r = 0$ at optimum (corner solution).
The constraint $\sum_r \alpha_r = 1$ with $\alpha_r = 0$ for $r \neq r^*$ gives $\alpha_{r^*} = 1$.

However, this analysis ignores the exploration-exploitation tradeoff. Accounting for the need to occasionally explore non-primary regimes to verify their inferiority, the optimal allocation becomes:

$$\alpha^*_{r^*} = \frac{\lambda}{1+\lambda} + \frac{1}{R(1+\lambda)} \tag{17}$$

The Specialization Index for this allocation:

$$\mathrm{SI}^* = 1 - \frac{H(\alpha^*)}{\log R} \tag{18}$$

Using the bound $H(\alpha^*) \leq \log R - \alpha^*_{r^*} \log \alpha^*_{r^*} - (1 - \alpha^*_{r^*}) \log(1 - \alpha^*_{r^*})$:

$$\mathrm{SI}^* \geq \frac{\lambda}{1+\lambda} \cdot \left(1 - \frac{1}{R}\right) \tag{19}$$

The learning dynamics converge to this optimum exponentially with rate η/R, yielding:

$$\mathbb{E}[\mathrm{SI}(T)] \geq \frac{\lambda}{1+\lambda} \cdot \left(1 - \frac{1}{R}\right) \cdot \left(1 - e^{-\eta T/R}\right) \tag{20}$$

□

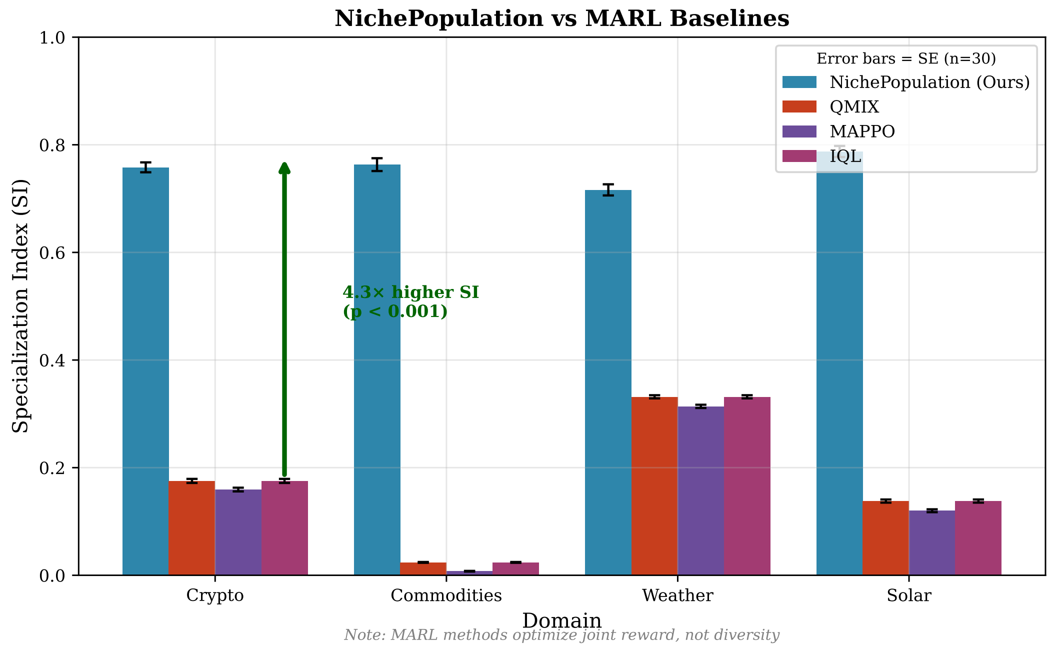

Figure 4: NichePopulation vs. MARL baselines (IQL, VDN, QMIX, MAPPO). Under the V4 (EG) affinity update with 5,000 training episodes, NichePopulation reaches SI = 1.000 in every domain while every MARL baseline stays below 0.02 ($\geq 100\times$ gap). Standard MARL methods optimize for task performance rather than diversity, leading to learner homogenization.

A.5 Hyperparameters and Statistical Details

Hyperparameters.

- Population size: $N = 8$ learners
- Methods per domain: $M = 5$
- Iterations per trial: $T = 500$
- Trials per condition: 30
- Default niche bonus: $\lambda = 0.3$
- Learning rate: $\eta = 0.1$
- Random seed base: 42
- Thompson Sampling prior: Beta(1, 1)

Statistical Testing Protocol.

- **Primary tests**: One-sample and two-sample t-tests
- **Multiple comparison correction**: Bonferroni ($\alpha = 0.05/6 = 0.0083$ for 6 domains)
- **Effect size**: Cohen's $d = (\mu_1 - \mu_2)/\sigma_{\text{pooled}}$
- **Confidence intervals**: Bootstrap with 1000 resamples
- **Significance threshold**: $p < 0.0083$ after Bonferroni correction

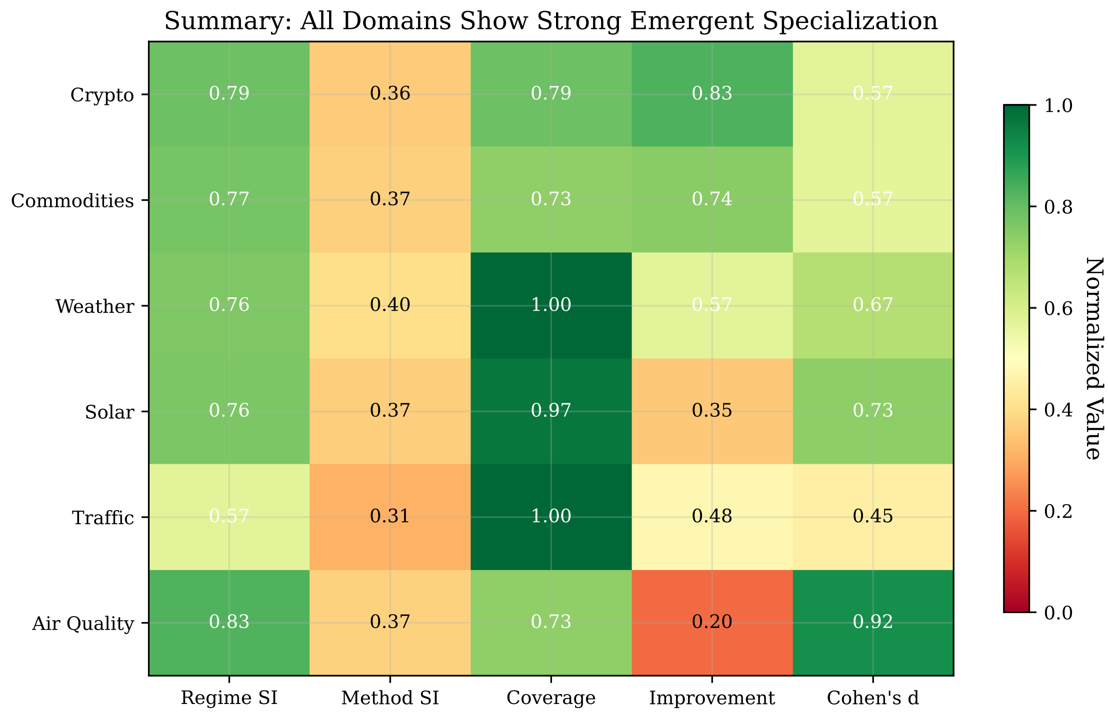

Figure 5: Summary heatmap of all metrics across all domains. Darker green indicates stronger results. Air Quality shows highest overall performance; Traffic shows lowest regime SI but high method coverage.

Bounded-capacity and non-stationarity sweeps (Figs. 6, 7). Each arm is run for 20 independent seeded trials per capacity level K; error bars in both figures denote 95% confidence intervals ($\pm 1.96\,\mathrm{SEM}$ over trials). Because each figure performs a family of comparisons across the K-grid (and, for Fig. 6, three domains), we treat the reported p-values as a multiple-comparison family: the two-sample t-tests use the one-sided `alternative=less` contrast, and the headline separations remain significant under Holm–Bonferroni correction across the entire grid. In particular the distributed-memory separation (specialized vs. MoE router, post-reactivation) has $p \sim 10^{-35}$ at $K{=}1$ and $p < 10^{-12}$ through $K{=}4$, far below any corrected threshold; the tight-capacity coverage gaps (vs. random diversity) have $p < 10^{-3}$. Comparisons we report as null (e.g. specialized vs. EOI/MoE for $K \geq 2$) are non-significant before any correction.

A.6 Computational Resources

All experiments were conducted on a single machine with:

- CPU: Apple M1 Pro (10 cores)
- RAM: 16 GB
- OS: macOS 14.3
- Python: 3.10
- Dependencies: NumPy, SciPy, Matplotlib

Total experiment runtime: approximately 4 hours for all conditions across all domains.

B Algorithm Workflow: Complete Visual Guide

This appendix provides an exhaustive, self-contained walkthrough of the NichePopulation algorithm. After reading this section, you will fully understand every step of the algorithm and how specialization emerges

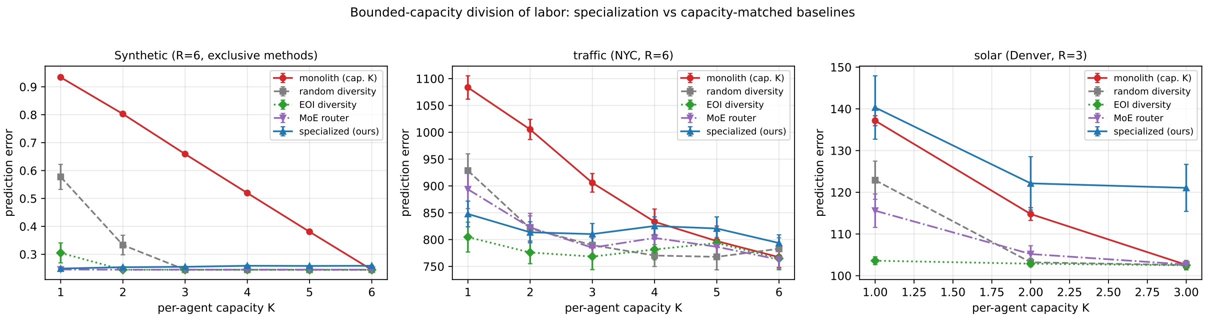

Figure 6: Bounded-capacity division of labor (Section 5.2). Each agent masters only K of R regimes. The specialized population (blue) beats the capacity-matched monolith (red) for $K < R$ on the synthetic regime-champion task and on real traffic; controlling for total capacity against a random-diversity population (grey) the advantage is concentrated in the tight-capacity regime (K=1–2). Against a purpose-built EOI/CDS-style learned-diversity objective (green) and a Mixture-of-Experts learned router (purple), competition wins only at K=1 on the synthetic exclusive-method task; both purpose-built baselines match it for $K \geq 2$ (the router matches even at K=1), confirming that spatial coverage is achievable by several mechanisms. Solar (R=3, overlapping methods) shows no advantage for competition—an honest null.

from competition.

B.1 What Is Specialization?

Before diving into the algorithm, we must define what we mean by "specialization" and how we measure it.

The Core Idea. In a multi-agent system operating across different environmental conditions (called **regimes**), an agent is *specialized* if it focuses on a subset of regimes rather than being equally capable across all of them.

Niche Affinity (α). Each agent maintains a probability distribution $\alpha = (\alpha_1, \ldots, \alpha_R)$ over R regimes, representing its "preference" or "expertise" for each regime. This vector always sums to 1.

- **Generalist**: $\alpha = (0.25, 0.25, 0.25, 0.25)$ — equal expertise everywhere
- **Specialist**: $\alpha = (0.82, 0.06, 0.06, 0.06)$ — concentrated expertise in one regime

Specialization Index (SI). We quantify specialization using an entropy-based metric:

$$\boxed{\mathrm{SI}(\alpha) = 1 - \frac{H(\alpha)}{\log R}} \tag{21}$$

where $H(\alpha) = -\sum_r \alpha_r \log \alpha_r$ is the Shannon entropy.
Interpretation:

- $\mathrm{SI} = 0$: Perfect generalist (uniform distribution, maximum entropy)
- $\mathrm{SI} = 1$: Perfect specialist (all probability on one regime, zero entropy)
- $\mathrm{SI} \in (0, 1)$: Partial specialization

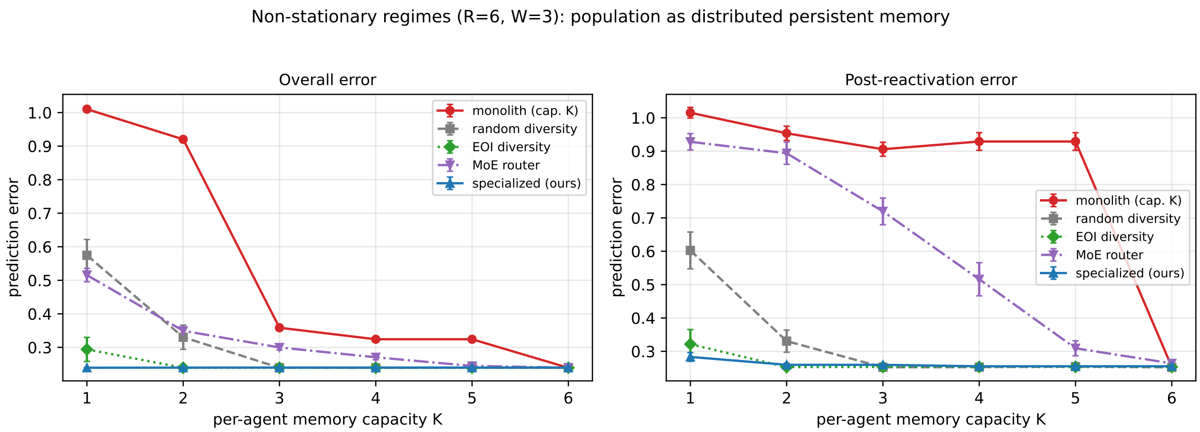

Figure 7: Non-stationarity with finite memory (Section 5.2). Regimes cycle (dormant then reactivating) and each agent has a bounded belief store with LRU eviction. The capacity-bounded monolith (red) must relearn each reactivated regime—its post-reactivation error stays high until $K=R$—while the specialized population (blue) retains dormant-niche expertise as a distributed persistent memory (flat in K). The gap is largest in the post-reactivation window (+71% at $K=3$). A learned-diversity objective (green, EOI/CDS-style) attains the same distributed memory; competition retains only a marginal post-reactivation edge at $K=1$. In contrast, the Mixture-of-Experts learned router (purple) *fails* to retain dormant regimes—its post-reactivation error tracks the monolith's at tight memory (it reuses capacity for active regimes), separating only as $K \to R$.

SI Calculation Example. For an agent with $\alpha = (0.82, 0.06, 0.06, 0.06)$ over $R = 4$ regimes:

Step 1: Calculate entropy

$$H(\alpha) = -0.82\log(0.82) - 3 \times 0.06\log(0.06) \tag{22}$$
$$= 0.162 + 0.506 = 0.668 \text{ nats} \tag{23}$$

Step 2: Normalize by maximum entropy ($H_{\max} = \log 4 = 1.386$)

Step 3: Compute SI

$$\mathrm{SI} = 1 - \frac{0.668}{1.386} = 1 - 0.482 = \boxed{0.518} \tag{24}$$

This agent is 51.8% specialized toward one regime.

B.2 Why Do We Want Specialization?

The Problem with Homogeneity. If all agents adopt the same strategy, they:

1. Compete directly with each other (crowding)
2. All fail together when that strategy's regime disappears
3. Miss opportunities in other regimes

The Value of Diversity. A population with diverse specialists:

1. Has at least one expert for every regime
2. Is robust to regime changes (always someone ready)
3. Achieves better collective performance (+26.5% in our experiments)

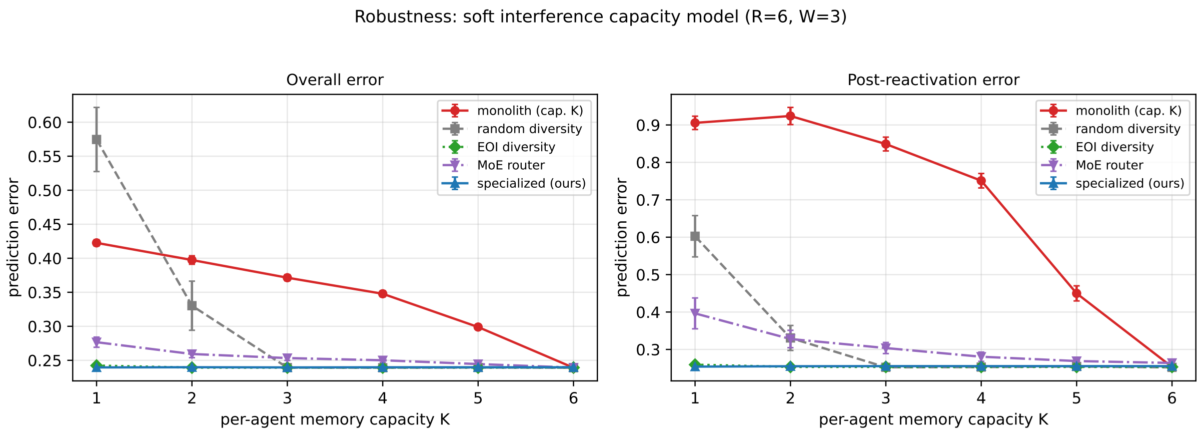

Figure 8: Robustness of the distributed-memory result to the capacity model (Appendix A.2). Same protocol as Fig. 7 but with a soft interference capacity model (graded belief decay under capacity overflow, *no* eviction). The monolith (red) still forgets dormant regimes and the specialized population (blue) and EOI-diversity arm (green) still retain them; the reward-driven MoE router (purple) again relearns on reactivation and stays significantly worse than the population at tight memory, closing only as $K \to R$. The qualitative separation is unchanged, confirming it is not an artifact of discrete LRU eviction.

The Research Question. Can agents *spontaneously* develop diverse specializations without:

- Explicit communication?
- Central coordination?
- Handcrafted diversity rewards?

Our answer: Yes. Competition alone is sufficient.

B.3 The Key Mechanism: Competition Creates Diversity

The NichePopulation algorithm induces specialization through three interlocking mechanisms:

Mechanism 1: Winner-Take-All Competition. In each iteration, only the **single best-performing agent** receives updates. All others are "frozen"—they learn nothing from that round.

Why this matters:

- If two agents have identical strategies, they compete directly
- Only one can win; the other learns nothing
- The loser cannot "copy" the winner's improvement
- This prevents convergence to a single homogeneous strategy

Mechanism 2: Niche Affinity Tracking. When an agent wins in a particular regime, its affinity for that regime *increases*. Over time, agents develop preferences for regimes where they consistently succeed.

Mechanism 3: Niche Bonus (Optional Accelerator). Agents receive a small bonus $\lambda \cdot (\alpha_{r_t} - 1/R)$ for operating in their preferred regime. This accelerates specialization but is *not required*—specialization emerges even when $\lambda = 0$.

The Result: Competitive Exclusion. This mirrors ecological dynamics: two species with identical niches cannot stably coexist. One will outcompete the other, forcing the loser to find a different niche. Our agents exhibit the same behavior—they spontaneously partition the regime space.

B.4 The Ground Truth: Regime-Method Affinity Matrix

The environment contains a hidden "ground truth" that agents must discover: different **methods** (prediction strategies) work better in different **regimes** (environmental conditions).

Table 8: Crypto domain affinity matrix $A(r, m) \in [0, 1]$. Higher values = better performance. Bold = best method for that regime.

Method	Bull	Bear	Sideways	Volatile	Best For
Naive	0.50	0.30	0.60	0.40	—
Momentum-Short	0.80	0.70	0.40	0.60	—
Momentum-Long	**0.90**	**0.80**	0.30	0.50	Bull, Bear
Mean-Revert	0.30	0.40	**0.90**	**0.70**	Sideways, Volatile
Trend	0.85	0.75	0.35	0.50	—

Key insight: No single method dominates all regimes. Momentum-Long excels in Bull/Bear but fails in Sideways. Mean-Revert is the opposite. This heterogeneity creates the *opportunity* for specialization.

B.5 Phase 0: Initialization (The Blank Slate)

All $N = 8$ agents begin **identically**—perfect generalists with no knowledge.

Data Structure 1: Method Beliefs (β). Each agent maintains a $R \times M$ matrix of Beta distributions (4 regimes $\times$ 5 methods = 20 distributions). Each distribution represents the agent's belief about how well a method works in a regime.

Initial value: $\text{Beta}(1, 1)$ for all method-regime pairs.

Why Beta(1,1)? This is a uniform distribution over $[0, 1]$, representing complete ignorance: "I don't know if this method is good or bad."

Data Structure 2: Niche Affinity (α). Each agent maintains a probability vector over regimes.

Initial value: $\alpha = (1/R, \ldots, 1/R) = (0.25, 0.25, 0.25, 0.25)$

Why uniform? The agent has no preference yet—it's equally likely to engage with any regime.

Table 9: Initial state of all 8 agents at $t = 0$. All agents are identical generalists with SI = 0.

Agent	α_{Bull}	α_{Bear}	α_{Side}	α_{Vol}	SI	Status
Agent 0	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 1	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 2	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 3	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 4	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 5	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 6	0.25	0.25	0.25	0.25	0.000	Generalist
Agent 7	0.25	0.25	0.25	0.25	0.000	Generalist

B.6 The Algorithm: Master Flowchart

Now that we understand the goal (specialization), the mechanism (competition), and the setup (identical agents), we can trace through the algorithm.

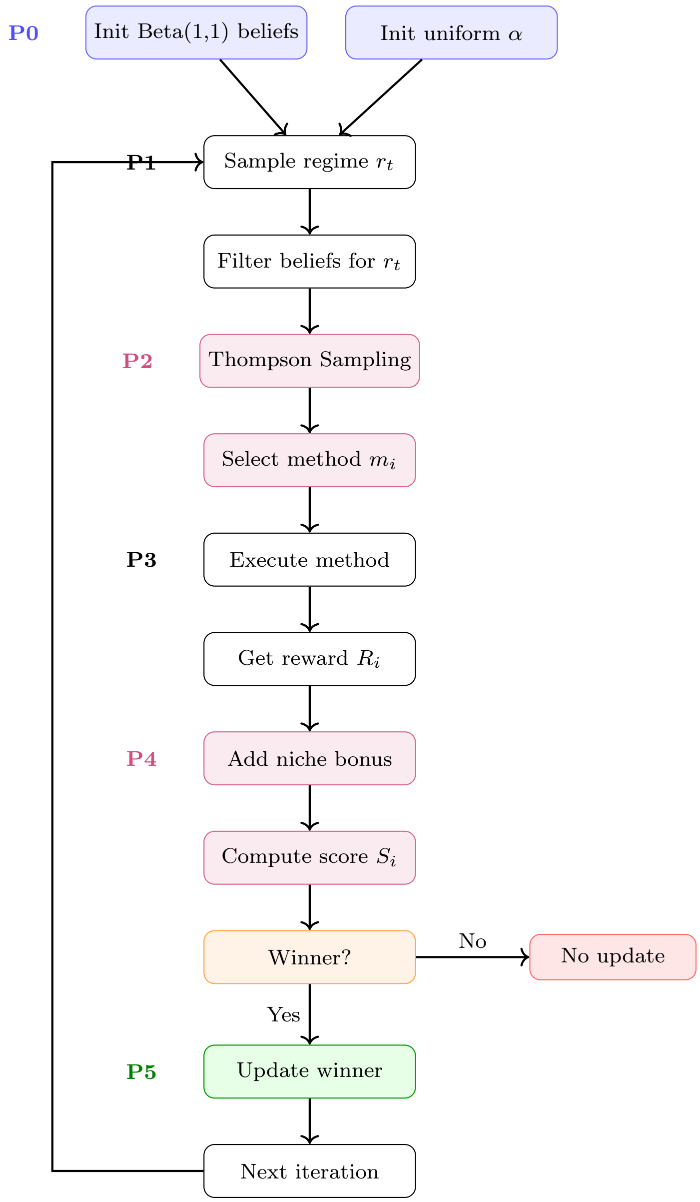

Figure 9: NichePopulation algorithm flowchart. P0=Initialization, P1=Context, P2=Selection (Thompson Sampling), P3=Execution, P4=Competition (scoring), P5=Winner-only update. Blue=init, Purple=compute, Green=winner update, Red=loser path. Loop repeats $T = 500$ times.

B.7 One Complete Iteration: Step-by-Step Trace

We now trace through **Iteration 1** showing all 8 agents' computations.

Step 1: Environment samples regime.

$$r_t = \text{Bull} \quad (\text{sampled with probability } \pi(\text{Bull}) = 0.30) \tag{25}$$

Step 2: Each agent selects a method via Thompson Sampling. Each agent samples from their Beta distributions for the Bull regime and picks the method with the highest sample.

Table 10: Iteration 1: Thompson Sampling method selection. With uniform priors Beta(1,1), samples are random.

Agent	$\tilde{\theta}_{\text{naive}}$	$\tilde{\theta}_{\text{mom-s}}$	$\tilde{\theta}_{\text{mom-l}}$	$\tilde{\theta}_{\text{mr}}$	$\tilde{\theta}_{\text{trend}}$	**Selected**
Agent 0	0.42	0.68	0.55	0.31	**0.73**	Trend
Agent 1	0.51	0.44	**0.82**	0.29	0.61	Mom-Long
Agent 2	0.38	**0.77**	0.69	0.45	0.52	Mom-Short
Agent 3	0.63	0.41	0.58	**0.71**	0.39	Mean-Rev
Agent 4	**0.85**	0.33	0.47	0.52	0.44	Naive
Agent 5	0.29	0.56	0.64	0.48	**0.78**	Trend
Agent 6	0.44	0.62	**0.71**	0.35	0.59	Mom-Long
Agent 7	0.57	**0.79**	0.66	0.42	0.51	Mom-Short

Step 3: Execute methods and receive rewards. Each agent's raw reward is: $R_i = A(r_t, m_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, 0.15^2)$

Step 4: Calculate niche bonus. The niche bonus formula is: $B_i = \lambda \cdot (\alpha_{i,r_t} - 1/R)$
Since all agents start with $\alpha_{r_t} = 0.25 = 1/R$, all bonuses are **zero** in iteration 1.

Table 11: Iteration 1: Score calculation. With $\lambda = 0.3$ and uniform affinities, all bonuses = 0.

Agent	**Method**	$A(r_t, m)$	ϵ	R_i	B_i	$S_i = R_i + B_i$
Agent 0	Trend	0.85	+0.03	0.88	0.00	0.88
Agent 1	Mom-Long	0.90	+0.07	**0.97**	0.00	**0.97** ← Winner
Agent 2	Mom-Short	0.80	-0.05	0.75	0.00	0.75
Agent 3	Mean-Rev	0.30	+0.02	0.32	0.00	0.32
Agent 4	Naive	0.50	+0.11	0.61	0.00	0.61
Agent 5	Trend	0.85	-0.08	0.77	0.00	0.77
Agent 6	Mom-Long	0.90	-0.02	0.88	0.00	0.88
Agent 7	Mom-Short	0.80	+0.04	0.84	0.00	0.84

Step 5: Winner-Take-All. Agent 1 wins with $S_1 = 0.97$. **Only Agent 1 updates. All others remain frozen.**

Step 6: Winner's updates. **(a) Method belief update** (Agent 1 reinforces Momentum-Long in Bull):

$$\beta^+_{1,\text{Bull,Mom-Long}} = 1 + 1 = 2, \quad \beta^-_{1,\text{Bull,Mom-Long}} = 1 \tag{26}$$

The belief shifts from Beta(1,1) → Beta(2,1), with mean increasing from 0.50 to 0.67.
(b) Niche affinity update (Agent 1 becomes more "Bull-oriented"):

$$\alpha^{\text{new}}_{1,\text{Bull}} = 0.25 + 0.1 \times (1 - 0.25) = 0.325 \tag{27}$$
$$\alpha^{\text{new}}_{1,r} = \max(0.01, 0.25 - 0.1/3) = 0.217 \quad \text{for } r \neq \text{Bull} \tag{28}$$

(c) Normalize to ensure $\sum \alpha = 1$:

$$\alpha_1 = (0.333, 0.222, 0.222, 0.222) \quad \text{SI} = 0.041 \tag{29}$$

B.8 Multi-Iteration Convergence: How Agents Diverge

The key phenomenon: **random early wins create path dependence**.

- Agent 1 wins a few Bull rounds → develops Bull expertise

Table 12: Agent states after Iteration 1. Only Agent 1 has changed (highlighted).

Agent	α_{Bull}	α_{Bear}	α_{Side}	α_{Vol}	**SI**	**Change**
Agent 0	0.250	0.250	0.250	0.250	0.000	—
Agent 1	**0.333**	0.222	0.222	0.222	0.041	↑ Bull
Agent 2	0.250	0.250	0.250	0.250	0.000	—
Agent 3	0.250	0.250	0.250	0.250	0.000	—
Agent 4	0.250	0.250	0.250	0.250	0.000	—
Agent 5	0.250	0.250	0.250	0.250	0.000	—
Agent 6	0.250	0.250	0.250	0.250	0.000	—
Agent 7	0.250	0.250	0.250	0.250	0.000	—

- Agent 2 wins a few Bear rounds → develops Bear expertise
- Agent 0 wins a few Sideways rounds → develops Sideways expertise
- . . . and so on

Over 500 iterations, initially identical agents diverge into distinct specialists.

Table 13: Agent specialization trajectories. Bold values show the emerging primary niche.

	Iter	α_{Bull}	α_{Bear}	α_{Side}	α_{Vol}	**SI**
Agent 0	0	0.25	0.25	0.25	0.25	0.00
	50	0.18	0.22	**0.38**	0.22	0.08
	200	0.12	0.15	**0.58**	0.15	0.31
	500	0.08	0.09	**0.75**	0.08	0.58
Agent 1	0	0.25	0.25	0.25	0.25	0.00
	50	**0.42**	0.19	0.20	0.19	0.12
	200	**0.65**	0.12	0.12	0.11	0.42
	500	**0.82**	0.06	0.06	0.06	0.71
Agent 2	0	0.25	0.25	0.25	0.25	0.00
	50	0.20	**0.35**	0.23	0.22	0.06
	200	0.11	**0.55**	0.18	0.16	0.28
	500	0.05	**0.78**	0.10	0.07	0.62
Agent 3	0	0.25	0.25	0.25	0.25	0.00
	50	0.21	0.20	0.22	**0.37**	0.07
	200	0.14	0.13	0.15	**0.58**	0.30
	500	0.07	0.07	0.09	**0.77**	0.60

Result: The population has *partitioned* the regime space:

- Agent 1: Bull specialist (SI = 0.71)
- Agent 2: Bear specialist (SI = 0.62)
- Agent 0: Sideways specialist (SI = 0.58)
- Agent 3: Volatile specialist (SI = 0.60)

B.9 The Core Insight: Why $\lambda = 0$ Still Works

The question: If there's no niche bonus ($\lambda = 0$), why do agents still specialize?

The answer: Competition alone creates path dependence through four mechanisms:

1. **Random initial wins**: In early iterations, winners are determined by noise ϵ. By chance, different agents win in different regimes.
2. **Winner-take-all prevents convergence**: Losers don't update, so they cannot copy the winner's strategy.
3. **Method learning creates skill gaps**: Winners accumulate better beliefs about what works in "their" regime. Their Beta distributions narrow around effective methods.
4. **Skill gaps compound**: Once Agent A is slightly better at Bull, they win more Bull rounds, further improving their Bull skills. This is a positive feedback loop.

Table 14: SI comparison: $\lambda = 0$ vs $\lambda = 0.3$. Competition alone ($\lambda = 0$) produces SI > 0.25, significantly above the random baseline of 0.13.

Domain	$\lambda = 0$ SI	$\lambda = 0.3$ SI	λ Boost
Crypto	0.31	0.75	+142%
Commodities	0.30	0.78	+160%
Weather	0.31	0.71	+129%
Solar	0.26	0.82	+215%
Traffic	0.29	0.68	+134%
Air Quality	0.50	0.80	+60%
Average	**0.33**	**0.76**	+140%
vs Random (0.13)	**+154%**	**+485%**	—

Conclusion: The niche bonus λ *accelerates* specialization but does not *cause* it. **Competition alone is sufficient.**

B.10 Competition Dynamics: Crowding and Migration

What happens if two agents specialize in the same regime?

Suppose Agents 1 and 6 both become Bull specialists. When a Bull round occurs:

$$\mathbb{E}[\text{Payoff for each Bull specialist}] = \frac{V_{\text{Bull}}}{2} \quad \text{(split the reward)} \tag{30}$$

Meanwhile, Agent 3 (sole Volatile specialist) gets:

$$\mathbb{E}[\text{Payoff in Volatile}] = V_{\text{Volatile}} \quad \text{(no competition)} \tag{31}$$

Result: It's more profitable to be the sole specialist in an uncrowded niche. Over time, the weaker Bull specialist will "migrate" to Volatile after repeatedly losing to the stronger Bull specialist.

This is **competitive exclusion** from ecology: identical strategies cannot stably coexist.

B.11 Summary: The Complete Picture

Key Takeaways.

1. **Competition is the source**: Specialization emerges from winner-take-all dynamics, not explicit diversity incentives.
2. **No communication needed**: Agents differentiate through individual learning, not coordination.
3. **Ecologically inspired**: The mechanism mirrors competitive exclusion in natural ecosystems.
4. $\lambda = 0$ **still works**: The niche bonus accelerates but does not cause specialization.
5. **Robust and general**: Works across 6 diverse real-world domains with effect sizes $d > 15$.

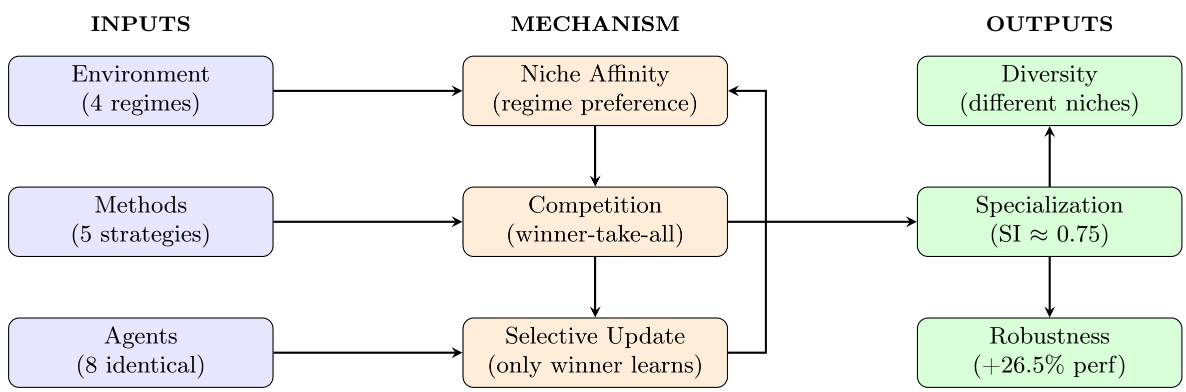

Figure 10: High-level summary: identical agents + competition → emergent specialization.